The Substratum Construction: Reconstruction, the Substratum Gauge Group, and the QM-GR Synthesis

Observer geometry and physical carrier. SM §4.1 distinguishes two claims. First, finite equivariant projection is classical and exact: cubic symmetry is inherited by the projected Markov part and every memory kernel, forbidding quadratic anisotropy in the scalar symbol — so whatever nondegenerate quadratic structure survives is isotropic — while allowing cubic anisotropy at quartic order. The particular normalized nearest-neighbor operator $A/(2d)$ additionally requires an **observer-level** center-free kernel; microscopic center independence alone does not imply that after correlated hidden variables are marginalized. Second, the six signed simple-cubic links and their exact $3 \oplus 2 \oplus 1$ representation are unconditional geometry, but identifying that link space with the complete physical carrier is **H-link**; the single-copy clause gives $K = 6$, and the custodial condensate-stabilizer reading is **H-cust**. Accordingly, any downstream statement here that consumes $K = 6$, $N_f = 6$, the physical SM gauge carrier, or a fixed three-sector count inherits those named bridge conditions unless it is explicitly referring only to the six geometric links.

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Abstract

Quantum mechanics and general relativity are conventionally treated as two fundamental theories awaiting unification. This paper develops the alternative that they are two projections of a single finite deterministic construction, joined by a third projection that produces the arrow of time. The underlying object is the bijection (S, φ) on a finite configuration space. Its visible-sector reduction has the exact finite stochastic/quantum-representation equivalence established in [Main]: finite observable laws, finite reversible realizations, and fixed-basis unitary/Born representations coincide, while genuine history readback forces hidden predictive memory. Its spatial coupling graph supplies finite causal cones; equality with the complete standard local quantum instrument theory is the narrower operational-lifting statement explicitly scoped in [Main]. The boundary-thermodynamic projection yields the gravitational description developed in [GR], and the lattice dynamics yields the Standard-Model structure developed in [SM].

Theorem 23 collects conditional local lattice/gauge outputs of a selected cubic branch under its named physical and structural premises. It does not

select that branch uniquely from E1–E7 and A1–A6; a Bell-compatible common completion additionally requires H-Bell. Theorem 24 establishes exact hidden-only coherent-channel invariance, broader rooted-population invariance and independent decoupled enlargement, together with separately scoped alphabet and graph comparisons. Pair C refutes exhaustiveness of the restricted coherent-channel generators, while equivalence across all admissible partitions remains open. These results motivate a common-substratum interpretation of the quantum, lattice-gauge and horizon constructions. They do not establish a complete observational quotient or a complete projection of its transformations onto the Standard Model gauge group.

1. Introduction

The standard physical picture of the world contains two fundamental theories. Quantum mechanics describes the microscopic regime in terms of unitary evolution on a Hilbert space, superposition, and Born-rule probabilities. General relativity describes the gravitational regime in terms of a deterministic, classical, dynamical metric obeying Einstein's equations. The two are mathematically and conceptually incompatible. Quantum mechanics has no preferred temporal foliation and no observer-independent state; general relativity has both. Quantum mechanics treats measurement as a primitive operation; general relativity treats it as a process within the dynamics. Quantum mechanics is linear; general relativity is not. Every attempt to unify them — to quantize gravity, to geometrize quantum mechanics, to merge the two into a deeper underlying theory — has produced either inconsistencies, ambiguities, or empirically unfalsifiable conjectures. The unification problem has resisted solution for nearly a century, and the most active current programs (loop quantum gravity, string theory, causal dynamical triangulations, asymptotic safety) have produced neither a quantitative empirical confirmation nor a clean conceptual resolution.

This paper argues that the QM-GR incompatibility is not a technical problem to be solved by finding the right mathematical framework. It is a category error: an attempt to merge two descriptions that turn out to be projections of the same underlying object viewed from different perspectives, rather than two competing accounts of the same regime. The framework developed in this four-paper synthesis to which this paper belongs — [Main], [SM], [GR], and the present paper — provides the underlying object and the projection map. The underlying object is a finite deterministic bijection (S, φ) on a configuration space partitioned into a visible sector accessible to an embedded observer and a hidden sector beyond the observer's causal reach. The projection from the underlying object to the observer's reduced description is the trace-out over the hidden sector. [Main] proves the exact finite observable-law equivalence $S \iff D \iff Q_{\mathrm{fb}}$: finite stochastic laws, finite reversible realizations, and fixed-basis unitary/Born representations coincide. The representation itself is universal; the memory-bearing OI sector is singled out instead by history readback, which forces hidden predictive memory. For spatial systems the same con-

struction supplies graph causal cones for finite-range couplings. What remains conditional on the quantum side is the strongest operational reading — one common standard quantum instrument/composite theory for the whole coherent intervention family. Bell is separate: the response-complete deterministic layer keeps measurement independence and reproduces quantum Bell violation only on the ontically parameter-dependent branch of [Main §3.3], with operational no-signaling. The projection from the same underlying object to general relativity is the thermodynamic limit at the partition boundary, formalized in [GR] as the cosmological-horizon application: applied to the cosmological horizon as a causal partition, the framework relates the action scale to the discreteness scale through the calibration $\hbar = c^3\epsilon^2/(4G\sigma)$, which fixes only $\epsilon/\sqrt{\sigma} = 2\,l_p$ for an entropy σ per boundary mode, and recovers the Bekenstein-Hawking entropy with the $1/4$ coefficient as a consistency relation of that calibration, both conditional on H-slope [GR §3.2], and proposes a dissolution of the cosmological constant problem conditional on the covariant coupling step [GR §6.2]. The Standard Model emerges from the same bijection on a cubic lattice with the wave equation as substratum dynamics: the cubic lattice is selected by the observed gauge group (the SU(3) factor requires the cubic crystal system), and — given that lattice — three generations (physical reading conditional on H-spin', [SM §4.7]; chirality on H-χ', [SM §4.8]), the Higgs as composite, and the $(3, 2, 1)$ gauge structure follow as forced consequences [SM]; strong CP is open there (H-top, H-det). The framework's hierarchical structural realism and its relationship to other unification programs are developed in the fifth core paper [Structure].

The present paper compares the visible-law construction of [Main], the selected lattice realization of [SM] and the horizon construction of [GR]. Theorem 23 (§3) states conditional outputs of the nearest-neighbor cubic wave-rule branch under H-link, the octahedral representation, amplitude-scale gauge and the separate physical hypotheses used by [SM] and [GR]. The converse from the empirical inputs to a unique local branch is unproved, even within A1–A6 without further selection assumptions. H-Bell separately conditions a common Bell-compatible completion. Section 4 distinguishes exact coherent-channel invariances from population-only invariance and structural alphabet/graph comparisons. The restricted coherent-channel generators are not exhaustive; the all-partitions classification and a complete Standard Model gauge-image map remain open. The proposed synthesis shares a model and observation machinery while retaining each construction's additional premises.

The synthesis proposed in §5 treats the quantum-representable visible law, the selected cubic gauge construction and horizon thermodynamics as descriptions associated with a common candidate substratum. [Main] supplies the exact finite observable-law representation. The physical Standard Model interpretation retains the hypotheses of [SM], while the action-scale and gravitational conclusions retain H-slope, the horizon/frame premises and the coupling conditions of [GR]. Sharing a candidate object does not prove that these descriptions constitute a unique common physical realization. Theorem 23 does not select that realization from observations, and Theorem 24 does not furnish a complete

gauge quotient. The horizon-growth account of the observer's arrows (§5.4) likewise retains its cosmological and observer-admission conditions.

The paper is organized as follows. §1.1 states the framework's primary metaphysical commitment and its relation to explanatory and empirical claims. §2 develops the physical interpretation of (S, φ) as a finite lossless memory with partial read-write access, and addresses the substrate objection (what is the memory made of?) by distinguishing demonstrated observational redundancies from the open question of deeper physical structure (§4). §3 states and proves the reconstruction theorem (Theorem 23). §4 proves selected fixed-interface invariances (Theorem 24), gives the restricted block-word rigidity result (Proposition 24.2), and examines the proposed gauge hierarchy with its open completeness question. §5 makes the synthesis claim explicit, citing [Main], [SM], and [GR] for the empirical content. §6 discusses structural realism, the ontological hierarchy, and the dissolution of the measurement problem. §7 concludes. The paper is short by design: the technical content fits in two sections (§§3–4) and the rest is the structural and interpretive context that shows why those two sections matter.

1.1 The metaphysical commitment

The framework's primary metaphysical proposal is ontic structural realism about the selected local model and the structures preserved by demonstrated observational redundancies. Theorem 23 supplies conditional outputs of that branch; it does not classify all microscopic representatives compatible with observed physics. Section 4 identifies invariances only for specified observational carriers. H-Bell is a separate compatibility condition for a common Bell-violating completion, and neither Theorem 23 nor Proposition 24.2 establishes its uniqueness. Commitment to one complete microscopic ontology therefore requires more than these results.

Stated at its most distilled, the framework's foundational commitment — what [Main §1.2] gives as "observation occurs" — is itself structured: it is two axioms. The first, that tokened differentiation occurs, is indubitable and conceptually primitive. The second, that differentiation recurs — across more than one domain, minimally configuration and time — is a substantive posit: not indubitable, not derivable from the first (a no-go result establishes this), though presupposing it. The commitment is single in the sense that the two axioms invoke one principle; it is, in count, two, and the second is the framework's one non-evidential foundational posit. Part I of the companion paper *Physics Modulo Gauge* itemizes what the first axiom forces as lemmas and what the second adds. (One item of that itemization is a reading of the primitive, adopted here explicitly: the observer's embeddedness — the proper-part decomposition $V \subsetneq S$ — belongs to the first axiom, because "tokened differentiation" is read perspectivally, as a registering from a locus against a remainder. On a thin reading it would instead be a third axiom; the framework adopts the perspectival reading, so the count is two. The choice is presentational, not empirical, and changes nothing downstream of (S, φ, V).)

The structural-realism commitment operates at multiple levels. Within OI's proposed structural class, the realism about $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$ depends on a specified observable equivalence and an unresolved classification of the transformations preserving it. Across structural classes — at the level of partial-trace observational features that any embedded-observer system must respect — broader structural realism applies, with the framework's universality-class equivalence ([Structure §9]) being the appropriate gauge equivalence. The full multi-level structure is articulated in [Structure], which canonicalizes both the observation hierarchy and the gauge hierarchy and their orthogonality, and develops the cross-framework analysis through universality classes of embedded observers. The substratum-class commitment of this paper is one level of a broader hierarchical structural realism: realism at multiple combinations of observation-hierarchy depth and gauge-hierarchy breadth, with the framework's empirical content concentrated at the deepest observation level (Level D in [Structure §2]) and the most class-specific gauge level (Level G3 in [Structure §2]). The articulation here focuses on this concentrated level because that is where the reconstruction theorem and substratum gauge group operate; the broader hierarchical content is developed in [Structure].

The structural-realist proposal has explanatory and empirical aspects. The exact finite-law representation in [Main §3] explains how a selected reversible model produces a quantum representation of its accessible law. Within the selected physical branch, the representation calculations and their named hypotheses constrain lattice/gauge outputs and the quantities classified in [SM §7] and [GR §7]. Their predictive or retrodictive status is that of the individual derivation chains. Neither explanatory economy nor agreement with those quantities supplies the missing converse from observations to a unique substratum. The structural, explanatory and empirical questions must therefore be assessed separately.

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2. The Physical Interpretation of (S, φ)

Storage and memory. S is the set of all distinguishable states: *finite capacity*. φ is a bijection: *perfect memory* — information is never created or destroyed. Together, (S, φ) is a finite lossless memory. The partition V defines the observer's *access* — both read and write. The observer reads the visible sector and, through the coupling H_{int}, visible operations write correlations into the hidden sector. The slow-bath form of (C2) is one mechanism that preserves those writes until readback. C4 then makes the visible law genuinely history-sensitive; on a fixed finite recurrent representative any non-permutation rooted map, which that readback supplies, forces indivisibility somewhere in the recurrence cycle. The resulting finite observable law has an exact fixed-basis unitary/Born representation by the foundational equivalence theorem. The unresolved step is not the existence or quadratic form of that representation, but its simultaneous extension to the full coherent local instrument/composite algebra. Interference in

the mechanism picture is write-then-read: information deposited in the hidden sector during one transition is retrieved later and changes the visible transition statistics.

The 10^{122} CC discrepancy is the compression ratio between total storage and readable storage. The dark sector is the gravitational effect of the unreadable storage. The Bekenstein-Hawking entropy is the storage capacity of the partition boundary. The action scale $\hbar$ is the conversion factor between storage geometry and read-write statistics.

Remark (the domain of the local lattice dynamics). The wave equation of [SM §4.1] — the unique second-order reversible nearest-neighbor dynamics compatible with center independence, isotropy, and linearity — governs the full **local lattice sector** used by the SM/GR construction, not the visible degrees of freedom alone. Visible and hidden lattice degrees of freedom in that sector run the same wave equation; the partition V imposes observer access without changing the local update. This is enough for (i) the trace-out of [Main] over lattice-hidden degrees of freedom, (ii) the nested local trace-out of [GR §8.4], and (iii) the link-carrier construction of [SM §4.7.1.2]. It is not enough for the Bell-violating deterministic completion under measurement independence: a nearest-neighbor update, whose cone advances one edge per update, together with interventions each reading and writing only its own wing and readouts outside the remote setting's graph cone, falls under [Main §3.3]'s Bell ceiling and gives $|S_{\mathrm{CHSH}}| \leq 2$. The one-object synthesis therefore additionally requires H-Bell (§3.3): the state-dependent graph must realize Bell-relevant preparation-indexed parameter dependence with operational no-signaling while preserving the metric/Ollivier–Ricci continuum limit required by the GR projection.

The substrate objection. "What is the memory made of?" A specified (S, φ), observer partition and preparation/readout rule determine a model's accessible law. Theorem 23 gives conditional local outputs of a selected realization, not a proof that this model is a complete description of reality or is uniquely recovered from observations. If two interpretations assign exactly the same accessible data to that model, any measurement procedure depending only on those data assigns them the same result, by the factoring argument [GR §6.3]. This conditional indistinguishability does not establish that every proposed microscopic difference is gauge. Questions about a deeper physical realization remain distinct from interpretive readings that stipulate identical data.

Relation to computation. A Turing machine has a tape (storage), a head (partial read/write access), and a transition function (update rule). The correspondence is suggestive: S is the tape, V is the head's access window, φ is the transition function. But three differences are physically significant. First, a Turing machine's tape is potentially infinite; S is finite — finiteness is essential for the recurrence proof of P-indivisibility, which holds on every finite reversible representative with a non-deterministic visible step; the representation itself is universal, though the effective finiteness result ([GR, §8.4]) shows the deep hidden sector may be infinite without affecting the leading-order transition data

on accessible timescales. Second, a Turing machine is generally irreversible (it can erase, overwrite, and halt); φ is a bijection — nothing is erased, nothing is created, there is no halting. Third, a Turing machine computes an *extrinsic function* (input $\to$ output for an external user); (S, φ) computes no extrinsic function — it is a closed permutation that cycles through states and returns. The appearance of dynamics, probability, particles, and forces is entirely the observer's perspective — what the permutation looks like through the partial window V.

Remark (the Turing connection under effective finiteness). The three differences soften under the effective finiteness result. With the deep hidden sector potentially infinite, the first difference is between the formal definition (S finite) and the physical requirement for accessible-timescale content (only $\mathcal{C}_V \times \mathcal{C}_B$ need be finite). The second difference is a specialization, not an opposition: reversible Turing machines (Bennett, Fredkin-Toffoli) are a well-studied subclass. The third difference — extrinsic vs. intrinsic — is the one that does physical work. The mapping is then structural: V is the head, H is the tape, φ is a reversible transition function, and C1–C4 characterize the architecture. The framework extends Turing's question: instead of asking what a machine can compute for an external observer, it asks what computation looks like to a component of the machine — and proves the answer admits the quantum representation universally — the memory-bearing sector, where the distinctly quantum phenomenology lives, characterized by C1/C3/C4.

The arrow of time. The substratum has no arrow of time — φ and φ^{-1} are equally valid. Observers satisfying C1–C4 are nonetheless structurally restricted to the non-equilibrium phase of (S, φ): the theorem in [Main, §4.6] shows, under its effective-mixing hypothesis and for the coarse-grained description generated by iterating the one-step effective channel, that $\tau_B(V)$ is bounded by the local mixing time on equilibrium microstates, so C2 fails there wherever that mixing time is short compared with τ_S — and hence C4, by the erasure lemma of [Main §2.3], which is the condition the characterization registers. This replaces the low-entropy "past hypothesis" with a structural observer-selection rule and excludes Boltzmann brains structurally rather than anthropically. Entropy increase is then emergent as a coarse-grained description of the non-equilibrium dynamics observers necessarily see.

Access, inference and computation. A proper instantaneous readout and a retained sequence are different observational carriers. Gödel's and Turing's undecidability results concern their own formal/computational hypotheses; they do not imply that a known finite reversible model has an unidentifiable hidden state. Finite record fibres and finite future-trace equivalence can be computed by enumeration or partition refinement. The finite-law representation and hidden-predictive-memory theorems do not require every hidden distinction to remain unknown.

Hidden-state inference. Fix known dynamics, an admissible initial support and the records the observer may retain. A hidden quantity is identified precisely

when it is constant on the states compatible with those records [Proposition 24.5]. For example, $\varphi(v, h) = (h, v)$ with readout v has $(X_0, X_1) = (v_0, h_0)$: two retained readings identify the initial pair and the current hidden bit $H_1 = X_0$. With a uniform initial law it nevertheless has $X_2 = X_0$ and positive history readback. C4 therefore need not leave residual hidden-state uncertainty after the full record. By contrast, an exactly decoupled spectator can remain in a non-singleton record fibre. Cardinality comparisons across different realizations are another question: Theorem 24(iii) preserves its specified visible channels under normalized independent decoupled enlargement; alphabet comparisons retain their stated hypotheses. Neither comparison proves universal state opacity.

Mathematics and physics. The trace-out performs a Jordan-Chevalley projection ([SM, Appendix A]): it extracts the semisimple part of the dynamics and erases the nilpotent monodromy. Physics is the semisimple shadow of mathematics — the diagonalizable spectral data, projected by the trace-out and organized by the gauge group's representation structure. Theorem 23 below gives conditional local lattice/gauge outputs on the selected cubic branch. It does not establish an inverse reconstruction of that branch or a complete observational gauge classification; Bell-inclusive existence and uniqueness impose further obligations.

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3. The Reconstruction Theorem

The forward construction associates a specified reversible model and observer interface with accessible laws [Main; SM §§3.1, 4–5]. The converse question is which model features observations identify. The three stages below organize possible temporal realizations and the additional premises of a selected spatial branch. Theorem 23 aggregates conditional outputs of that branch. It does not prove a unique inverse within A1–A6, and H-Bell separately conditions a common Bell-compatible completion.

3.1 Inputs to the reconstruction

The reconstruction takes two kinds of inputs: empirical observations and structural assumptions. Being explicit about both is essential to the uniqueness claim.

Empirical inputs (facts about the observed universe):

(E1) **Unitary quantum mechanics.** The observed physics is described, to the precision tested, by quantum mechanics: states as vectors in a complex Hilbert space, unitary time evolution, self-adjoint observables, Born-rule outcome statistics.

(E2) **Bell violations.** The observed correlations violate Bell inequalities, ruling out local hidden-variable theories with factorizable distributions.

(E3) **Finite boundary entropy.** The entropy of any bounded region is finite and scales as the area of its boundary. This is a theoretical input — the holographic bounds [1, 2], which are not themselves observed — applied to the cosmological horizon, whose finite area makes the bound apply.

(E4) **Spatial isotropy.** Observed physics is rotationally invariant; no spatial direction is preferred.

(E5) **Propagating gravity.** Gravitational degrees of freedom propagate (gravitational waves are observed). This requires $d \geq 3$, since the Weyl tensor vanishes for $d \leq 2$. (Used in Stage 2(a) below; one of the three independent forward filters that select $d = 3$.)

(E6) **Stable matter.** Atoms, planets, and bound states exist on observed timescales. This requires $d \leq 3$, since the Coulomb potential in d dimensions gives unstable atoms for $d \geq 4$ (Ehrenfest [10]) and gravitational orbits are unstable for $d \geq 4$ (Tangherlini [11]). Without stable matter, no embedded observers exist to define the partition. (Used in Stage 2(a); second forward filter for $d = 3$.)

(E7) **Boundary-entropy concordance.** The cosmological parameter inferred from observation under the standard cosmological model satisfies $\rho_s/\rho_{\text{crit}} \approx 1$ (spatial flatness near critical density); it is an inferred quantity, not a directly measured one. In d spatial dimensions, the general-d form of this ratio is $\rho_s/\rho_{\text{crit}} = 2/(d-1)$, which equals unity only for $d = 3$. (Used in Stage 2(a); third forward filter for $d = 3$. See [SM §3.2] and [GR §7.2] for the derivation.)

Structural assumptions (restrictions on the class of candidate substrates (S, φ)):

(A1) **Finiteness.** The configuration space S is finite. (Two-part status. E3, with the holographic bound read as a Hilbert-space dimension cutoff $\dim \mathcal{H} \leq e^{A/4}$, bounds what observation reaches: the visible sector and its boundary layer, $\mathcal{C}_V \times \mathcal{C}_B$ — on which the emergent observables depend at leading order on accessible timescales, by the boundary-only dependence lemma [GR §3.2]. For a deep hidden sector that no dynamics couples to the boundary layer, the extension beyond the observer's access is observationally redundant for the specified visible channel, rather than an established full all-partitions gauge identification: by the deep-sector enlargement freedom (Theorem 24(iii)), such a sector may be finite of any size, or infinite, without changing any transition probability, and A1 selects the minimal — finite — representative of the equivalence class. A deep sector coupled to the boundary layer with slow dynamics changes the transition probabilities at $\mathcal{O}(t/\tau_B)$; its size is not resolved by the leading-order transition law on accessible timescales, which is the scope of the transfer in the corollary following Theorem 24. In the decoupled case the substrate's total cardinality beyond the boundary layer is not an observable question, in the same sense that the uniform vacuum offset is not.)

(A2) **Determinism.** $\varphi : S \to S$ is a bijection (deterministic, reversible dynam-

ics). (Two-part status, per the dilemma argument of [Main §2.2]: a non-injective descent either leaves a statistical trace — excluded where the observed statistics are of the doubly stochastic class that the uniform-prior marginal of a bijection produces — or leaves none, in which case it is removable, every map on a finite set restricting to a bijection on its eventual image, where all recurrent registered history lives. Bijectivity for the dynamics governing registered history is thus partly structural and partly anchored by one empirical input, the observed statistics in that scope; Main's honest-accounting ledger records the same division.) The anchor is empirically live: its current frontier is nanoparticle matter-wave interference at macroscopicity $\mu = 15.5$ — sodium clusters above 1.7×10^5 amu held in 133-nm path superpositions [4] — an order of magnitude beyond the prior benchmark, and the direct one-sided test against the intrinsic-collapse alternatives that would break it.

(A3) **Bounded coupling degree.** Each site is coupled to a bounded number of neighbors in φ's action. This restricts the graph class used for locality and dimensional structure. The graph comparisons of Theorem 24(iv) retain their stated hypotheses and structural conclusions; they do not establish equality of complete channel families for every bounded-degree graph in a quasi-isometry class.

(A4) **Center independence.** The dynamics φ does not depend on a choice of "center" site, in two senses used below: φ commutes with lattice translations up to gauge, and the update at a site contains no explicit copy of that site's present value ([SM §4.1]). (The second sense is what the wave-equation derivation of Stage 2 uses. Its observational anchor is the homogeneity of physical law — no experiment has found a preferred location in the laws — the cosmological principle's complement to E4's isotropy; whether it is independently warranted by the observer architecture remains the question flagged in §1.)

(A5) **Linearity.** The wave equation for φ is linear. (Equivalent to amplitude-scale gauge invariance — that the field-value scale is unphysical — via the linearity-equivalence lemma of [SM §4.1]; hence necessary *given* that gauge principle and "partly derived" in exactly that sense. Whether the q-size gauge freedom of [SM §2.7] entails amplitude-scale gauge is an identified open step; pending it, nonlinear alternatives are excluded precisely to the extent amplitude-scale gauge is assumed, and would otherwise require a separate derivation.)

(A6) **Background independence.** The dynamics is covariant under spatially-varying internal-index transformations: a site-dependent transformation $G(\mathbf{n})$ of the internal index is a symmetry of the dynamics provided the coupling data carried on the links are transformed with it, $M(\mathbf{n}, \hat{e}_j) \to G(\mathbf{n})\, M(\mathbf{n}, \hat{e}_j)\, G(\mathbf{n} + \hat{e}_j)^{-1}$ ([SM §3.1]). The content of the assumption is this covariant interface — the coupling is link-valued data on which the transformations act — and once it is in place the local transformation law imposes no further condition on the rule. A6 is not the stronger fixed-background condition that the same transformations leave the dynamics invariant with the coupling held fixed and

preserved pointwise: for an invertible constant coupling that condition forces the transformation to agree across every coupled pair of sites, and at the symmetric point of the coupling matrix it fails for every transformation that differs across a coupled pair. The global commutant symmetry — the transformations constant across the lattice that leave the coupling itself unchanged, the specialization of A6 in which nothing needs transporting — is what the local gauge reading of [SM §3.1] proceeds from. The state-dependent coupling graph of [SM §3.1], under which the graph itself evolves with the state, is a separate principle that shares the name and is not A6.

Theorem 23's branch-specific conclusions use E1–E7 and A1–A6 together with additional model-selection premises named in §3.3; E1–E7 and A1–A6 alone are not shown to identify a unique substratum. The set A1–A6 is used in the conditional local construction but is not proved sufficient to select a unique realization, nor proved minimal: A4 (center independence) and A6 (background independence) overlap in physical content (A6 promotes the symmetry that A4 constrains), and A5 (linearity) is equivalent to amplitude-scale gauge invariance via the linearity-equivalence lemma of [SM §4.1], hence partly derived — necessary *given* amplitude-scale gauge, with whether [SM §2.7]'s q-size gauge entails it left as an identified open step. A tighter axiomatization may be possible but is not pursued here; the stated local consequences also depend on the explicit branch choices and completion premises, not only on the independence of A1–A6. The empirical inputs E5, E6, E7 (propagating gravity, stable matter, boundary-entropy concordance) provide explicit forward filters for $d = 3$ — see Stage 2(a) below.

Critical dependencies. Theorem 23's proof relies on the following prior theorems, whose individual correctness is assumed:

- [Main §2.3] history readback for the **memory-bearing temporal** sector used in Stage 1 (a modeling premise about the temporal data, not an inference from Bell violation)
- [Main §3.3] Bell ceiling and the adopted measurement-independent, ontically parameter-dependent Bell branch (a separate spatial-composite premise; Bell violation is not used as a witness of temporal P-indivisibility)
- [Main §3.2] Forward Stinespring dilation (bijection → CPTP channel) and reverse-direction lemma (one-step doubly stochastic matrix → uniform-prior bijection; exact for rational entries, to arbitrary precision otherwise)
- [Main §3.4] Finite-horizon process dilation (stochastic process → bijection, per finite horizon)
- [Main §3.4] Characterization theorem (accessible non-Markovianity — the framework's finite-horizon visible-law memory notion, not generic quantum non-Markovianity — $\iff$ C1/C3/C4 per finite accessible horizon; C2's slow-bath necessity conditional on the hidden-sector mixing hypothesis, ETH its physical motivation, its persistence form holding without it)
- [SM §3.2] Coupling-graph dimension → d = 3

- [SM §4.1] Center independence + isotropy + linearity → wave equation
- [SM Theorems 5–15] Gauge group, generations, hypercharges derivation chain
- [SM Theorems 17–21] T-invariance → $\theta \in \{0, \pi\}$ and reciprocal transition counts; $\bar{\theta}$ open (H-top, H-det)
- [GR §3] Gap equation from thermal self-consistency → $\hbar = c^3\epsilon^2/(4G\sigma)$

Theorem 23's branch-relative conclusions inherit the stated limitations of these dependencies and of the additional assumptions in §3.3.

3.2 Stage 1: Observed QM → deterministic embedding with C1–C4

Inputs: E1 (unitary QM), E2 (Bell violations), E3 (finite boundary entropy), A1 (finiteness), A2 (determinism), and M1, in two clauses: **M1-T**, the observed temporal quantum data used in Stage 1 lie in the accessible history-readback sector of [Main §2.3–§3.4]; and **M1-B**, Bell-violating composites take [Main §3.3]'s measurement-independent, ontically parameter-dependent branch with operational no-signaling. M1-T and M1-B are logically distinct; Bell violation is not taken to imply temporal P-indivisibility.

Output: There exists a triple (S, φ, V) with S finite, φ a bijection on S, and $V \subset S$ a distinguished subset such that: - The observed quantum dynamics is the reduced description obtained by tracing out $S \setminus V$ from the deterministic evolution under φ. - The triple satisfies C1 (non-trivial coupling), C2 (record persistence — the slow bath one realization), C3 (sufficient hidden memory capacity), and C4 (history readback) for the M1-T temporal sector: [Main §3.4]'s universal hidden-memory theorem — accessible non-Markovianity $\Longleftrightarrow$ C1/C3/C4 realization per accessible horizon, with $M_t > 0$ forcing unavoidable hidden predictive memory — makes readback-relevant hidden memory unavoidable in every faithful deterministic realization. M1-B separately requires Bell-relevant cross-wing coupling/parameter dependence and does not supply C4.

Conditional structure. C1, C3, and C4 for the memory-bearing temporal sector are derived from E1–E3, A1–A2 together with **M1-T** — C4 via the hidden-memory theorem applied to the stipulated accessible temporal memory; conditional on M1-T, they hold without further hypothesis. **M1-B** is separate: it maps E2 to the ontically parameter-dependent Bell composite and supplies an independent cross-wing-coupling requirement, not C4. C2 takes the two forms of [Main §1.3]. (C2-structural), retention of the record of visible history to its readback, holds by reversibility in every bijective realization. C2's necessity holds in its persistence form without further hypothesis: in the coupling-step model of [Main §3.4], observed conditional memory $I^* > 0$ places some realized hidden conditional at the coupling time at least $\varepsilon_{\min}(I^*, n_V)$ from its reference prior (the physical-memory theorem). Its slow-bath form is conditional on the quantitative mixing hypothesis of [Main §3.4] — uniform total-variation closeness of the hidden sector's post-relaxation conditionals to a single bath distribution — under which the same bound makes $\varepsilon_{\min}(I^*, n_V)$ a lower bound

on the relaxation residue at the coupling time, hence on the bath's conditional mixing time, and a departure δ from the homogeneous Markov benchmark at step order k forces $\varepsilon \geq \delta/(k-1)$ (the bound $\|T^{(k)} - T^k\|_{\mathrm{TV}} \leq (k-1)\varepsilon$, constant optimal at $k = 2$); without it, accessible conditional memory does not bound the bath's mixing time. ETH is the physical case for the hypothesis, as discussed in [Main §3.4]'s "Remark (Status of ETH)." ETH is a well-supported conjecture for generic chaotic many-body Hamiltonians and is the standard assumption for the cosmological-horizon complement; relaxations or failures of ETH would require a separate argument. The framework's gauge-group derivation through Stage 2 inherits this conditional.

Bell–gravity compatibility hypothesis (H-Bell). The Stage-2 reference nearest-neighbor wave-equation graph has a pointwise causal cone advancing one edge per update; whether that cone or the lift's long-wavelength speed $1/\sqrt{3}$ is the physical c is not derived [Main §3.3]. Under measurement independence, setting interventions that each read and write only their own wing and ± 1 readouts outside the remote setting's graph cone, [Main §3.3] therefore makes that **reference graph** Bell-local and caps CHSH at 2. The framework nevertheless already makes the graph state-dependent: the discrete Einstein equation of [SM §3.1] is formulated on $G(x)$, and its mod-q spatial-Markov proof explicitly assumes uniform initial conditions. M1-B may therefore be implemented by **preparation-indexed adjacency**: the vacuum/uniform reference class retains the geometric bounded-range graph used by the local SM/GR derivation, while an entangling preparation supplies bounded-degree Bell-relevant long edges (or an equivalent ontically nonlocal response map) furnishing parameter dependence without operational signaling.

H-Bell is the remaining compatibility statement for that branch: the preparation-indexed graphs must (i) supply the required ontic parameter dependence, (ii) preserve operational no-signaling, and (iii) satisfy a metric/Ollivier–Ricci stability condition strong enough that the continuum-curvature step of [SM §3.1] remains valid on the state classes to which the Einstein reconstruction is applied. The graph-boundary area law is not the primary unresolved issue: $|\partial V|$ is defined on $G(x)$ and therefore counts preparation-induced crossing edges. The open step is the identification of that graph boundary/curvature with continuum geometric area/Ricci curvature. [SM §3.1]'s sparse local-contamination estimate, $L_N m_N/N \to 0$, is a sufficient condition for typical mesoscopic neighborhoods to remain reference-like, but it does not prove global curvature convergence, control defect endpoints, or establish the integrated-curvature limit. H-Bell is therefore an explicit curvature-stability/composite-existence hypothesis, not a consequence of A1–A6 or of the nearest-neighbor wave equation.

Two points limit the exposure this conditional creates. First, the relevant question is not whether ETH holds universally — it is known to fail for integrable and many-body-localized (MBL) systems — but whether it holds for *the* hidden sector implied by the framework's other commitments. The hidden sector here

is the high-capacity complement ($N_H \gg N_V$ — the comfortable realization of C3's capacity floor) of a bounded-coupling bijection on a statistically isotropic graph (A3, E_isotropy). Integrability and MBL are both non-generic in a specific sense relevant here: integrability requires an extensive set of local conserved quantities, which a statistically isotropic bounded-degree coupling graph without fine-tuned commuting structure does not possess; MBL requires strong quenched disorder sustaining localization against the thermalizing tendency of generic interactions, and is moreover known to be unstable in dimension $d \geq 2$ (the avalanche instability). Since the reconstruction independently selects $d = 3$ (§3.3, [SM §3.2]), the hidden sector sits in the regime where MBL does not survive and generic interacting dynamics thermalize — so ETH is the expected case, not an additional fine-tuning. Second, the conditional attaches only to the slow-bath (timescale) form of C2's necessity direction; C2's persistence form and the sufficiency direction — a C1–C4 realization yields a visible law with conditional memory (forced by C4), and that finite law universally admits the quantum representation — are unconditional, so the framework's positive predictive content does not rest on ETH. The mixing hypothesis, for which ETH is the physical case, is used only to convert observed memory into a lower bound on the bath's conditional mixing time; the reconstruction's uses of that form hold within the class of hidden sectors satisfying the mixing hypothesis. ETH, which the argument above makes the expected case at $d = 3$, is the physical case for that hypothesis and does not by itself supply it: a hidden sector satisfying ETH for few-body observables while retaining conditional-distribution memory violates it ([Main §3.4], Remark (role of ETH)).

Derivation. The observed quantum dynamics provides, for each finite experimental context, a stochastic law on a finite visible configuration space. By [Main]'s exact finite-horizon equivalence, this law has a finite reversible deterministic realization and an exact fixed-basis unitary/Born representation. The representation step is universal and therefore does not itself reconstruct the nontrivial OI sector. The additional modeling premise M1 separates two empirical structures. **M1-T** places the temporal data used for Stage 1 in the accessible history-readback sector; [Main §2.3] then proves that on a fixed finite representative a non-permutation rooted map, which C4 readback supplies, forces indivisibility somewhere in the recurrence cycle. **M1-B** places the Bell data on [Main §3.3]'s measurement-independent, ontically parameter-dependent branch; Bell violation therefore certifies Bell-relevant cross-wing coupling but is not used to infer temporal P-indivisibility. By [Main]'s finite-horizon equivalence ($S \Rightarrow D$, the process-dilation and response-table constructions of [Main §3.4]), the temporal law admits a realization as the visible marginal of a deterministic bijection φ on an enlarged finite space $\mathcal{C}_V \times \mathcal{C}_H$, with a hidden prior independent of the visible initial state carried as part of the realization datum; for a single one-step doubly stochastic matrix the reverse-direction lemma ([Main §3.2]) gives the uniform-prior form, exactly when its entries are rational and to arbitrary precision otherwise. This delivers the triple (S, φ, V) with $S = \mathcal{C}_V \times \mathcal{C}_H$ and $V = \mathcal{C}_V$. Non-permutation $T(t)$ in the M1-T temporal sec-

tor gives the corresponding C1 witness; independently, M1-B's ontic parameter dependence requires Bell-relevant cross-wing coupling. The necessity analysis of [Main §3.4] establishes, for every faithful realization of the resulting dynamics, C3 (the data-processing bound $\log_2 |\mathcal{C}_H| \geq I^*$, with no mixing hypothesis) and (C2-structural) (retention of the record of visible history to its readback, supplied by reversibility; record persistence is required for the non-Markovian returns that produce quantum interference); in the coupling-step model of [Main §3.4], observed conditional memory keeps some realized hidden conditional at least $\varepsilon_{\min}(I^*, n_V)$ from the reference prior at the coupling time (C2's persistence form, with no mixing hypothesis), and under the hidden-sector mixing hypothesis ([Main §3.4]; ETH its physical motivation) this lower-bounds the bath's conditional mixing time (the slow-bath form).

Stage-1 dilation freedom. For the finite channel/dilation data used in temporal realization, hidden-basis freedom is defined at the level of an individual channel. Exact hidden-factor relabelling and independently decoupled finite enlargement preserve the full time-resolved channel under the hypotheses of Theorem 24. Neither fact classifies all dilations of a common channel family. This is not uniqueness of the full triple (S, φ, V) across all coherent composites. In particular, it does not select a preparation-indexed Bell adjacency rule, and the operational lift still requires its separately stated finite-predictive-dimension, local-tomography and purification hypotheses.

Status: Finite-horizon realization and its stated memory conditions; no unique dilation, selected spatial branch or Bell-inclusive completion follows from this stage alone.

3.3 Stage 2: Embedding + isotropy → specific lattice + gauge structure

Inputs: Stage 1 output + E4 (spatial isotropy) + E5 (propagating gravity) + E6 (stable matter) + E7 (boundary-entropy concordance) + A3 (bounded coupling) + A4 (center independence) + A5 (linearity) + A6 (background independence) + max-lattice-speed condition (for the wave-equation coupling constant; see step (b) below). Identification of this local lattice output with the Bell-violating Stage-1 completion additionally assumes H-Bell.

Output: The local propagating substratum sector is a $d = 3$ cubic lattice bijection with: - $K = 2d = 6$ internal components per site - Coupling matrix eigenvalue multiplicities $(3, 2, 1)$ - Gauge group SU(3) × SU(2) × U(1) - Three generations of chiral fermions in the SM representations - One Higgs doublet $(\mathbf{1}, \mathbf{2}, +1/2)$ - Anomaly-free hypercharges $(Y_Q, Y_u, Y_d, Y_L, Y_e) = (1/6, 2/3, -1/3, -1/2, -1)$

Derivation.

(a) *Dimension.* The coupling graph inherited from Stage 1 has polynomial growth exponent d. Three independent forward filters from the empiri-

cal inputs converge on $d = 3$: propagating gravity (E5) requires $d \geq 3$ (the Weyl tensor vanishes for $d \leq 2$); stable matter (E6) requires $d \leq 3$ (Coulomb instability of atoms for $d \geq 4$, plus Tangherlini's gravitational orbit instability); boundary-entropy concordance (E7) gives $\rho_s/\rho_{\text{crit}} = 2/(d-1) = 1$ at $d = 3$ uniquely. The intersection of the first two filters already pins $d = 3$; the third is a sharpening consistency check that the framework passes exactly. The integer-polynomial growth assumption (rather than fractal or exponential growth) is itself derived from self-consistency: exponential growth fails Lorentz-invariant dispersion (Jacobson's GR derivation requires it); fractal growth fails the cubic-lattice symmetry needed for the cubic-group decomposition in step (d) below. By Gromov's theorem and statistical isotropy (E4), the surviving graph class is quasi-isometric to $\mathbb{Z}^d$ for integer d; the three filters then select $d = 3$. ([SM §3.2] presents this argument in full; the inputs E5, E6, E7 surfaced here make explicit the empirical content the filters require.)

(b) *Wave equation.* Center independence (A4, in its functional sense: no self-term), isotropy (E4), and linearity (A5) uniquely select, within the nearest-neighbor class (an assumption beyond A3's bounded degree; see the scope remark at [SM §4.1]), the wave equation up to a multiplicative coupling constant α ([SM §4.1, Theorem]): the unique second-order linear dynamics on a lattice that is translation-invariant, isotropic, reversible and free of a self-term has the form $f = \alpha(x_1 + x_2 + \cdots + x_{2d}) \bmod q$ with coupling coefficient α. The coefficient is neither the propagation speed nor free to be set to 1: causal support travels at most one lattice edge per update whatever its value, and on the observer-level normalized branch it equals $1/d$ ([SM §4.1, Remark]).

(c) *Internal components.* H-link gives $K = 2dm$, with the additional single-copy clause $m = 1$ giving $K = 6$; coupling-degree minimization does not establish that clause [SM §4.7]. The selected staggered construction has a singlet/triplet candidate-sector decomposition. The identification of the triplet as three physical fermion generations retains H-spin' and the separate matter hypotheses; it does not follow from the link count alone.

(d) *Gauge group.* The six geometric links decompose as $T_1 \oplus E \oplus A_1$. Under H-link single-copy + H-cust, a generic equivariant condensate has unitary stabilizer $G_0 = U(3) \times U(2) \times U(1)$ [SM Theorems 5 and 7]. The pointwise commutant of the cubic action itself is $U(1)^3$, distinct from this stabilizer. A precise correlated two-torus quotient gives $G_0/K \cong S(U(3) \times U(2))$ [SM Proposition 14a], and the selected SM charged family has faithful image $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ [SM Proposition 15a]. Amplitude-scale invariance alone does not select K or that matter action; the full native endomorphism carrier has only the common scalar circle in its kernel. The physical action and remaining phase selection are additional premises. A6 supplies the separate covariance of site-dependent transformations with transformed link data, not the global quotient.

(e) *Matter content.* Staggered reduction gives three degenerate spin-1/2 sectors (Theorems 8–10). Fermion embedding is completed via the link-carrier construction ([SM §4.7.1.2]). Partition-spinor identification (Theorem 12) and trace-out (Theorem 13) give chirality. Anomaly cancellation (Theorems 14–15) uniquely fixes the hypercharges.

(f) *Discrete symmetries.* T-invariance of the bijection φ gives $\theta \in \{0, \pi\}$ and reciprocal visible transition counts at every scale (Theorems 17–21); it does not fix $\bar{\theta}$, which remains open under H-top and H-det [SM §5]. Reciprocity leaves the Jarlskog invariant free; only a mechanism making Y_u and Y_d simultaneously real would force $J = 0$, which is the falsification control on H-det proposals [SM, §5.3–5.5].

Scope of the Stage-2 selection. Steps (a)–(f) do not supply a chain of converse implications from the listed outputs to a unique substratum. The dimension filters use E5–E7; step (b) works within a nearest-neighbor, second-order, linear wave-rule class stronger than bounded coupling alone; step (c) uses H-link and the selected multiplicity $m = 1$; and the subsequent representation and matter conclusions use the cubic/octrahedral representation, amplitude-scale gauge and the qualifications in [SM]. These give conditional structural outputs on the selected branch. They neither exclude every other substratum satisfying E1–E7 and A1–A6 nor establish uniqueness modulo a fully characterized $\mathcal{G}_{\text{sub}}$.

Bell compatibility. Step (b)'s nearest-neighbor **reference** cone cannot itself realize M1-B, while retaining measurement independence, in a Bell protocol whose wings are separated by more than the graph cone over its duration. Stage 2 therefore establishes the local lattice/gauge sector on the geometric reference class. The state-dependent $G(x)$ machinery leaves a native route for M1-B through preparation-indexed adjacency, but the Ollivier–Ricci continuum theorem presently cited by [SM] does not cover those long-edge perturbations. Nor does it cover the unperturbed reference graph, and there the situation is worse than open: the degree-6 cubic hop metric is exactly ℓ_1, with a scale-independent $\sqrt{2}$ diagonal stretch, so the hop-metric curvature route fails for the reference family and no $D = 3$ extension of the cited theorem would rescue it. The repair is to re-found curvature on the propagation/Laplacian geometry, which the substratum's isotropic leading dispersion already supplies [SM §3.2]. Promoting the local sector and M1-B to one common (S, φ) is therefore conditional on H-Bell's curvature-stability requirement.

Status: Conditional results for the selected local lattice branch; uniqueness over all candidate substrata and a common Bell-inclusive completion remain open.

3.4 Stage 3: Dynamics → emergent fundamental constants

Inputs: Stage 2 output.

Output: The emergent action-scale relation $\hbar = c^3\epsilon^2/(4G\sigma)$, equivalently $\epsilon =$

$2\sqrt{\sigma}\, l_p$, where l_p is the Planck length and σ the entropy per boundary mode ([GR §3.1]); it fixes only $\epsilon/\sqrt{\sigma}$, not the value of $\hbar$ ([GR §4]).

Derivation. The gap equation from thermal self-consistency of the boundary layer ([GR §3]) gives the stated relation. The derivation uses: existence of thermal equilibrium at the boundary layer (a consequence of the C1–C4 dynamics from Stage 1), the boundary-only dependence lemma ([GR §3.2]), and the infrared detailed-balance condition H-slope with the horizon and frame conditions ([GR §2, §3.2, §8.5]), which does not follow from φ being a bijection ([GR §3.2]) and is an assumption beyond Stages 1–2.

Uniqueness at Stage 3. The gap equation admits a unique solution under the stated inputs ([GR §3.2]).

Status: Conditional on H-slope with the horizon and frame conditions ([GR §3.2]).

3.5 The composite theorem

Lemma 23.0 (Shared outputs on the selected local branch). *Suppose two finite reversible models both implement the same selected local branch: spatial dimension $d = 3$, a nearest-neighbor cubic wave-rule carrier, H-link with $m = 1$, the octahedral representation and the amplitude-scale-gauge and matter-sector hypotheses explicitly invoked in [SM]. Then they have the same values of the branch's discrete representation invariants $K = 6$ and multiplicities $(3, 2, 1)$, together with those gauge and matter outputs that follow from the cited [SM] results under their respective hypotheses. If both additionally satisfy H-slope and the horizon/frame assumptions of [GR §3.2], they obey the same action-scale relation $\hbar = c^3\epsilon^2/(4G\sigma)$; this does not fix a common numerical $\hbar$ without fixing ϵ and σ. Equality of these outputs does not imply the two substrata are related by any proposed gauge generator.*

Proof. Under the common selected branch and H-link with $m = 1$, $K = 2dm = 6$; the stipulated octahedral representation has the fixed decomposition $(3, 2, 1)$. The remaining stated gauge and matter outputs are inherited, only under their individual hypotheses, from the corresponding [SM] results. The [GR] calibration under H-slope supplies the relation among $\hbar$, ϵ and σ. None of these deductions identifies the full microscopic dynamics or reconstructs a unitary from the single-time visible channel; Theorem 24 establishes only its stated fixed-interface invariances and Proposition 24.2 has stronger block-word hypotheses. □

Theorem 23 (Conditional local-branch reconstruction). *Assume the finite reversible framework with the stated E1–E7 and A1–A6 inputs, together with a selected $d = 3$ nearest-neighbor cubic wave-rule realization, H-link at $m = 1$, the requisite octahedral representation, amplitude-scale gauge and the separate matter-sector premises in [SM]. On that selected branch the internal carrier count is $K = 6$ and its octahedral multiplicities are $(3, 2, 1)$; the stated gauge-group, matter, chirality and hypercharge outputs follow under their respective*

[SM] hypotheses, and the relation $\hbar = c^3\epsilon^2/(4G\sigma)$ follows only under H-slope and the horizon/frame assumptions in [GR]. The branch's reference local causal dynamics cannot by itself supply the Bell-violating M1-B composites under measurement independence; a common completion is conditional on H-Bell. The hypotheses do not establish uniqueness of the local branch among all candidate substrata, completeness of a substratum gauge group, or uniqueness of a Bell-inclusive completion.

Proof. The selected wave-rule and H-link premises give the internal-count and octahedral outputs by the local constructions of §3.3 and [SM]. The further gauge, matter and hypercharge statements use exactly the additional premises and derivation steps named in [SM]; they are not promoted to unconditional consequences of E1–E7 and A1–A6. Under the separate [GR] horizon conditions the calibration relation follows. Stage 1's finite dilation represents temporal laws; this alone supplies neither the selected spatial branch nor a Bell-compatible common realization. The reference nearest-neighbor causal graph is Bell-local over separated wings under the stated measurement-independence and readout conditions; the proposed preparation-indexed completion is subject to H-Bell and is not constructed here. Lemma 23.0 compares only the stipulated branch's outputs, and neither it nor Proposition 24.2 classifies all substrata modulo gauge. □

3.6 Remarks

Remark (Evidential weight of the construction). The selected cubic branch yields representation multiplicities $(3, 2, 1)$ under H-link with $m = 1$ and the stipulated octahedral action. Gauge and matter interpretations inherit their separate premises, including H-cust, H-spin' and H-χ' where invoked; the hypercharge result retains its family-pattern assumptions. [SM §7] records classified quantitative comparisons within that branch. Agreement with Standard Model data tests the applicable chains, but does not establish that A1–A6 and the empirical inputs select this branch uniquely. The physical branch selector and the observational equivalence classification require independent arguments.

Remark (Conditional content of the local construction). Theorem 23 states consequences of explicit premises. Once the $d = 3$ nearest-neighbor cubic wave-rule branch, H-link with $m = 1$ and its octahedral representation are fixed, $K = 6$ and the multiplicities $(3, 2, 1)$ follow. Further gauge, generation, chirality and hypercharge conclusions use the respective [SM] physical hypotheses; the action-scale relation additionally uses H-slope and the horizon/frame conditions of [GR]. These deductions constrain a chosen construction. They do not show that all models compatible with E1–E7 and A1–A6 share these structures or belong to one gauge orbit. Equality of the listed outputs also does not identify the full microscopic dynamics. The inverse selection problem remains open within A1–A6 without further premises.

Remark (Derived outputs, empirical selection and model premises). The repre-

sentation multiplicities follow from the selected link representation. The physical interpretation of the resulting blocks requires the separate carrier and matter hypotheses of [SM]. The empirical arguments for $d = 3$ and the cubic crystal system are conditional selection arguments in their stated classes; they do not classify every reversible model allowed by A1–A6. The nearest-neighbor wave rule, site factorization and single-copy H-link clause are additional premises. The derivation-status discussion in [SM §8.3] and the observable classifications in [SM §7] distinguish these roles. None supplies a theorem that observations recover a unique local branch or a complete microscopic gauge class.

The structural assumptions A1–A6 exclude several broad classes of theory from the outset:

- Intrinsically continuum theories (violate A1)
- Intrinsically stochastic theories (violate A2)
- Theories with unbounded coupling degree (violate A3)
- Theories violating center independence, linearity, or background independence (violate A4–A6)

Within the selected cubic branch, the specified local outputs follow under the hypotheses listed in Theorem 23. Uniqueness of all realizations modulo a complete observational gauge group is not established, and neither is Bell-inclusive substratum uniqueness. A separate argument would be required to rule out alternatives outside this class. The framework's defense of A1–A6 (in [SM §2] and elsewhere) is that they are natural given the empirical inputs E1–E7, but this defense is itself an argument, not a theorem.

Remark (Hypothesis dependencies). Each structural assumption can in principle be weakened or replaced:

- A1 (finiteness) is supported by E3; alternative would be an effective-finiteness argument with continuum UV completion, which the framework does not develop.
- A2 (determinism) is the key ontological commitment; stochastic alternatives would produce different reconstructions.
- A3 (bounded coupling) is conventional for physical systems; long-range couplings would give different dimensional structure.
- A4 (center independence) rules out preferred-frame theories.
- A5 (linearity) is the strongest restriction; it is equivalent to amplitude-scale gauge invariance (linearity-equivalence lemma, [SM §4.1]), so nonlinear wave equations on finite lattices are the separate class one obtains exactly when that gauge principle is dropped.
- A6 (background independence), as covariance under local internal-index transformations with the link coupling transformed alongside, is the standard gauge-theoretic requirement; the stronger fixed-background invariance, with the coupling held fixed, is a separate condition and is not what A6 asserts.

Weakening any A_i does not merely enlarge the equivalence class; it potentially

produces *different* reconstructions that may not agree with observation. The programme proposes that E1–E7 + A1–A6, together with the selected cubic branch and further conditions, form an explanatory model; a unique selection of that model and the exclusion of all alternatives have not been established.

Remark (Continuum extension). The lattice-level predictions are the framework's primary claims — the lattice is the fundamental description, not an approximation to a continuum theory. The structural results (gauge group, representations, generation count) are algebraic/topological properties of the lattice theory and are exact. Quantitative observables (scattering amplitudes, mass ratios) are compared to experiment through continuum perturbation theory; the lattice-continuum discrepancy at any experimentally accessible energy E is suppressed by $(E\epsilon/\hbar c)^2 = 4\sigma(E/M_{\mathrm{Pl}})^2$, which at $\sigma = 1$ is $\sim 10^{-32}$ at LHC energies. The rigorous proof that lattice Yang-Mills defines a continuum limit with a mass gap (a Clay Millennium Prize problem) is an open problem in mathematics, but it is not required by the OI framework: the lattice theory is complete, and its predictions are lattice-exact.

The epistemic structure is as follows. A reasonable objection is that the framework treats the lattice as fundamental (so the mass-gap problem can be set aside) while also using continuum perturbation theory to extract the numbers that match data (which would make the continuum limit load-bearing). The position is consistent. The framework's claim is that the lattice theory is the complete physical content and its predictions are exact *at the lattice level*. The comparison to experiment is then a comparison between two objects: the lattice theory's prediction extracted via continuum PT (a controlled approximation with error bounded by $(E\epsilon/\hbar c)^2 = 4\sigma(E/M_{\mathrm{Pl}})^2 \sim 10^{-32}$ at $\sigma = 1$), and the continuum-QFT fit to the experimental data (the standard way the data is reported). The empirical agreement is therefore agreement between the lattice theory's continuum approximation and the continuum-QFT data fit, with the approximation error in the first object ~ 30 orders of magnitude below the experimental precision of the second, so the lattice-vs-continuum gap is not the source of any observed agreement or disagreement at accessible energies. The mass-gap problem is not load-bearing for this: it concerns whether a continuum limit *exists as a rigorous mathematical object*, whereas the framework requires only that continuum PT is an accurate *calculational approximation* to the lattice theory's predictions at low energy — the standard status of lattice-to-continuum perturbative matching, which does not require the existence of the rigorous limit. The lattice is fundamental, the continuum is a calculational tool with a quantified error, and the experimental comparison is between the tool's output and continuum-reported data.

Remark (Finite observation sets vs. idealized empirical inputs). The empirical inputs E1–E7 are stated as structural facts about observed physics — the full apparatus of QM, Bell violations attaining Tsirelson's bound, the holographic entropy bound, exact rotational invariance, propagating gravitational waves, stable matter on all observed scales, and spatial-curvature concordance. Op-

erationally, no finite observation set fully establishes any of these; finite measurements support them only in the sense of being consistent with the data so far. Theorem 23 gives conditional outputs for the selected cubic branch even when the empirical inputs are idealized as exact; neither local-branch selection nor Bell-inclusive uniqueness follows from those idealized observations alone. Reconstruction from finite observation sets is a related but distinct question: with finite data, multiple equivalence classes may all be consistent, with the data discriminating among them only weakly. The framework's operational answer to the finite-data question is the cumulative weight of the quantitative predictions of [SM §7] — twenty-two classified retrodictions of gauge couplings, CKM/PMNS angles, mass ratios, and the Koide relation (two parameter-free structural and five conditional structural, the remainder layered, chained or fitted, the gauge couplings among the fitted), the measured comparisons each at $\lesssim 1\sigma$ — which collectively rule out reconstructions that would have produced different values. Theorem 23 states the branch-specific structural consequences, while [SM §7] records experimental comparisons for quantities computed or fitted within that branch; those comparisons alone do not prove uniqueness of its microscopic realization.

The implication supplied by Theorem 23 is conditional on the selected cubic realization and its named physical premises:

$$\text{Selected branch and hypotheses} \quad \Longrightarrow \quad \text{local outputs.}$$

The arrow does not give a converse from observations to the selected branch. A physical selector, an appropriate observational equivalence relation and a valid classification would be needed for a stronger inverse theorem; the restricted coherent-channel generators already fail to be exhaustive by Pair C. H-Bell separately conditions a common Bell-compatible completion, whose uniqueness is unproved. Interpreting a model's mathematics as reality or as a description produces the same measurements only when the interpretations stipulate the same accessible data. That factoring statement does not turn a complete ontology or a resolution of Wigner's puzzle into a consequence of Theorem 23 or Theorem 24.

Remark (ER=EPR at substratum and emergent levels). The Maldacena-Susskind ER=EPR conjecture [3] proposes that any two entangled systems are connected by an Einstein-Rosen bridge, with the strong form asserting that entanglement *is* a non-traversable wormhole — the same ontological structure under two descriptions. The framework here engages this conjecture at two distinct levels.

At the **emergent level**, the conjecture's *weak* form is reproduced: any pair of entangled subsystems in the emergent QM description shares boundary modes via the partition trace-out, and the holographic dictionary maps shared boundary modes to a connecting geometric structure in the emergent gravitational description. This recovers the ER=EPR weak form — entangled systems are

connected by a wormhole-like geometric object in the emergent theory — as a structural consequence of the framework's holographic architecture rather than as an independent postulate.

At the **substratum level**, the conjecture's *strong* form (entanglement is identical to wormhole geometry, not merely correlated with it) is predicted *false* at the representative level. Two distinct partition representatives can produce identical emergent EPR correlations while having different substratum representative structures — there is no faithful bijection from EPR pairs to substratum geometric features. The strong identity holds at most up to gauge equivalence on the substratum side. Whether a gauge-orbit reformulation of strong ER=EPR — "EPR-equivalence-classes correspond to ER-equivalence-classes" — is true in the framework remains open; the test would be whether the substratum's gauge group $\mathcal{G}_{\text{sub}}$ acts transitively on the set of substratum representatives that produce a given EPR correlation pattern.

This produces a sharp empirical signature: the framework's operational predictions agree with weak ER=EPR (which would also be predicted by AdS/CFT, by quantum error correction in holography, and by other emergent-geometry approaches) but disagree with strong ER=EPR at the representative level. Distinguishing these experimentally would require probing substratum-scale features, which is currently beyond reach. The disagreement is therefore a structural prediction rather than a testable one at present, but it concretely positions the framework relative to the broader ER=EPR program.

3.7 Companion-family identification

Within the specified finite companion family $\Phi_F(x, y) = (F(x) - y, x)$, a known labelled finite abelian group and exact complete current-field triple support, from a full-support phase ensemble, determine F. Under uniform phase counting the classical one-step population kernel is instead uniform for every F; this need not hold for a structured stationary prior. The companion identification note gives the proof, a complete-parent-patch version and a coarse-readout control. This restricted inverse assumes its readout and ensemble coverage; a single orbit need not supply them. It supplies no physical branch selector, and Theorem 23's inverse selection problem remains open.

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4. The Substratum Gauge Group

For a specified visible/hidden decomposition $S = V \times H$ and a specified hidden preparation ρ_H, the **fixed-interface coherent-channel equivalence** $U \sim_{\text{ch}} U'$ means equality, for every integer $t \geq 0$ and every visible input density matrix ρ, of the channels

$$\Phi_t^U(\rho) = \text{Tr}_H[U^t(\rho \otimes \rho_H)(U^\dagger)^t].$$

The weaker **population equivalence** $U \sim_{\mathrm{pop}} U'$ compares only $T_{ij}(t) = \langle i|\Phi_t^U(|j\rangle\langle j|)|i\rangle$ for every i, j, t, at the same interface and preparation. A third reading compares observables **across all admissible observer partitions**; it requires a specified correspondence between those partitions, their preparations and their readouts. No classification of that third relation is proved here. A simultaneous relabelling of the dynamics *and* all prepared/read-out observables is only a passive change of description; it cannot certify equivalence at a fixed interface. The expression $\mathcal{G}_{\mathrm{sub}}$ below denotes a **proposed** observational gauge structure relative to a stated observable carrier, not an already classified common gauge group for these three readings.

Theorem 24 (Exact interface invariances and their different scopes). *Let V, H be finite and let U be a permutation unitary on $\mathbb{C}^V \otimes \mathbb{C}^H$, with uniform hidden preparation. (i) A single hidden permutation $\sigma : H \to H$, applied identically in every visible fibre, preserves Φ_t for all t under conjugation by $\mathbb{1}_V \otimes P_\sigma$. More generally, a fibre-dependent hidden relabelling $\pi(v, h) = (v, \sigma_v(h))$ preserves all rooted transition probabilities $T_{ij}(t)$, but need not preserve coherent channels. (iii) Adjoining an exactly decoupled ancillary dynamics and an independently prepared normalized ancilla preserves Φ_t for all t wherever the tensor-product channel and partial trace are defined. Clauses (ii) and (iv) below are separate structural/conditional comparisons, not assertions of equality of all coherent channels. Restricted hidden-only relabelling and decoupled enlargement do **not** generate every fixed-interface coherent-channel equivalence (Pair C below). Completeness across all partitions remains open.*

(i) Relabellings and observation. For a single σ, its permutation unitary is $\mathbb{1}_V \otimes P_\sigma$, and the maximally mixed hidden state is invariant. The outer hidden conjugations disappear under the partial trace, so the coherent-channel identity holds for arbitrary visible density matrices. For $\pi(v, h) = (v, \sigma_v(h))$, the visible component of π is unchanged, and each rooted fibre's uniform hidden prior is permuted bijectively. Therefore the visible output counts, at every iterate, are unchanged. However the controlled hidden permutation generally acts differently on off-diagonal visible coherences and need not leave Φ_t invariant. An **arbitrary** permutation of all states need preserve neither data set when the readout is fixed. For example, take $V = H = \{0, 1\}$, $\varphi(v, h) = (h, v)$, and exchange $(0, 0)$ with $(1, 0)$ while fixing the other two states. At $t = 1$ the rooted transition matrices are

$$T_\varphi(1) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}, \qquad T_{\pi\varphi\pi^{-1}}(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

These disagree already on populations and thus on the coherent channels. If the readout and preparation are also transported by π, the agreement is a passive coordinate identity, not a new gauge invariance. A visible-layer swap T in a specific wave-rule branch need not be of hidden-only form and is **not** thereby a generator in the coherent-channel group; its possible time-reversal interpretation is a separate structural statement.

(ii) Alphabet size. The results in [SM §2.7] that hold for every $q \geq 2$ transfer **without** an extra alphabet-size hypothesis. Other quantities in that section retain explicit q dependence. Equality of transition probabilities across different alphabet sizes is a **separate hypothesis**, and the unconditional q-insensitive structural conclusions do not establish that hypothesis or equal coherent-channel families.

(iii) Decoupled ancillary enlargement. For a separate system D, a normalized ancillary density operator ρ_D, and unitary/bijective independent dynamics ψ, the enlarged evolution is $U^t \otimes \psi^t$. Its reduced visible channel obeys

$$\mathrm{Tr}_{H\otimes D}[(U^t \otimes \psi^t)(\rho_V \otimes \rho_H \otimes \rho_D)(U^{\dagger t} \otimes \psi^{\dagger t})] = \Phi_t^U(\rho_V)\,\mathrm{Tr}(\rho_D) = \Phi_t^U(\rho_V).$$

This is exact for finite D and also for infinite-dimensional D with well-defined unitary dynamics, normalized trace-class ρ_D and the indicated partial trace. It does not transfer finite-state recurrence to an infinite enlargement. A **coupled** deep sector, however slowly coupled, is not covered by this equality; the accessible-time approximation has the separate error term of [GR §3.2].

(iv) Structural graph comparisons. Theorems 1–6 and [SM §2.2] retain their **specified** structural conclusions within the graph classes and assumptions to which those derivations apply: polynomial-growth dimension, mode-count and statistical-isotropy conditions can support corresponding emergent-QM, dimension and gravitational *structural* content. This is stronger than simply noting common graph labels, but it is **not** an equality theorem for complete rooted channels. The Standard Model cubic decomposition additionally uses the octahedral point-group representation; generic quasi-isometry does not preserve it. Neither these structural results nor a graph isomorphism alone proves coherent-channel equivalence at a fixed observer interface.

Proof of the exact invariances. The coherent statement in (i) follows from conjugation by the single hidden unitary, invariance of the uniform hidden density, and the partial-trace identity. For the population statement in (i), each σ_v bijectively permutes the preparation fibre H and fixes the final visible coordinate; applying $\pi\varphi^t\pi^{-1}$ proves equality of every rooted count. Statement (iii) follows from tensor-product evolution, unitary preservation of the independent ancillary trace, and partial trace. The structural claims (ii) and (iv) retain only the indicated source hypotheses and conclusions. □

Completeness depends on the observable carrier. For **fixed-interface coherent channels** $\sim_{\mathrm{ch}}$, the proposed restricted hidden-only transformations and decoupled enlargements do **not** generate all equivalences: the six-state Pair C dynamics have equal Φ_t for every t but are separated by the balanced block-word trace invariant tested by `ST3_pairC` and `ST4_pairC`. The invariant survives hidden-only conjugation and, after the appropriate normalization, independent decoupled enlargement; its behavior under an allowed visible-coordinate relabelling is also included in the tested Pair C obstruction. Thus merely chaining these moves cannot connect the pair. This is a **restricted non-exhaustiveness result**, not a statement about arbitrary additional moves. For

population-only equivalence $\sim_{\text{pop}}$, the larger fibre-dependent relabellings preserve $T_{ij}(t)$ and the Pair C obstruction does not establish non-exhaustiveness under that relation. For **all-partitions** equivalence, the relevant observational information and permitted correspondences have not been settled, so completeness remains **open**. None of these three verdicts can be substituted for another.

Corollary (effective finiteness and gauge-class transfer). Three statements make A1's two-part status precise. *(i) The bound.* E3, read as a Hilbert-space dimension cutoff, bounds the observationally relevant space: $|\mathcal{C}_V \times \mathcal{C}_B| \leq e^{S_\partial}$ with $S_\partial = A/4$ in Planck units — for the de Sitter horizon, $e^{10^{122}}$. This dimension-cutoff reading is the interpretive premise of A1, and this is the only place it enters. *(ii) Boundary-only dependence.* For accessible times $t \ll \tau_B$, $T_{ij}(t) = T_{ij}^{(B)}(t) + \mathcal{O}(t/\tau_B)$ ([GR §3.2]), so at leading order every emergent observable on accessible timescales is a functional of $\mathcal{C}_V \times \mathcal{C}_B$ data alone. *(iii) Transfer.* Under a decoupled enlargement of the kind in Theorem 24(iii), with a normalized independent ancillary state and well-defined partial trace, every $T_{ij}(t)$ is unchanged at every t, whether the ancilla is finite or infinite-dimensional. Statements expressible solely through those probabilities transfer; finite-state recurrence and other statements not fixed by that law do not transfer to infinite systems. An extension coupled to the boundary layer with $\tau_B^D \gg \tau_S$ alters $\mathcal{C}_D$ and its dynamics while leaving $T^{(B)}$ unchanged, so the two systems share only the leading-order law: a statement expressible through $\{T_{ij}^{(B)}(t) : t \ll \tau_B\}$ transfers by (ii), while a statement about the transition probabilities themselves transfers only if it is insensitive to $\mathcal{O}(t/\tau_B)$ corrections. A transition probability equal to zero on one system and to $\varepsilon = \mathcal{O}(t/\tau_B) > 0$ on the other is the elementary statement that does not transfer. P-indivisibility at the recurrence scale — established on finite representatives by recurrence ($\varphi^N = \mathrm{id}$, [Main §2.3]) — is not an accessible-timescale statement: across coupled extensions it holds on the finite members, and on separately specified infinite-deep-sector models, where finite recurrence is unavailable, the readback lemma ([Main §2.3]) gives the accessible-window property it states, conditional memory, a different and weaker conclusion. A1's choice of the finite representative is therefore a convenience of proof for content carried by the leading-order accessible-timescale law; recurrence-scale content is stated for finite representatives. □

Remark (amplitude-scale gauge). The operational principle used here declares an overall field normalization redundant when preparations and readouts are compared under the corresponding alphabet-coordinate change. Affine changes $\phi \mapsto \lambda\phi + c$, with λ a unit, are available coordinate relabellings; observational invariance for a specified carrier is an additional requirement, not a conclusion about every fixed readout. The principle is separate from alphabet-size comparisons and from Theorem 24's hidden-only channel invariance. Its role in the conditional linearity argument of [SM §4.1] does not imply that independent block phases can all be removed while retaining hypercharge. The exact correlated-quotient construction and its additional physical-action premise are stated in [SM Proposition 14a].

Status (open, referee-grade). Amplitude-scale gauge is adjoined as an explicit operational principle: it does not follow from Theorem 24(i), which here allows only hidden-only channel relabellings (and, for population data, fibre-dependent hidden relabellings); clause (ii) concerns alphabet-*size* comparisons, not arbitrary field-value changes. Results invoking amplitude-scale gauge therefore remain conditional on that separate principle. Stated operationally it is *not* viciously circular: the input (absolute amplitude is unobservable) is strictly weaker than the output (additive dynamics), and a nonlinear F on the same alphabet is a coherent object the principle *excludes* rather than presupposes. The one route that would make linearity *unconditional* — deriving the unobservability of absolute amplitude from the observer architecture itself — is by contrast circular: what the embedded observer can resolve through the trace-out is dynamics-dependent (a nonlinear F induces amplitude-dependent emergent dispersion, which *is* observable), so "absolute amplitude is unobservable" already presupposes linearity. Amplitude-scale gauge therefore has the status of an operational input on a par with center independence and isotropy, not a theorem; whether it is independently warranted by the observer architecture is the question flagged for external review.

Formalization status of the linearity step (resolved as a sharpened axiom, not a closure). A dedicated audit of this step settles its logical status, and the outcome is worth stating precisely because it is the prototype for how the framework's "open" foundational steps resolve. The linearity-equivalence lemma — *all first finite-differences of the update rule over $\mathbb{Z}/q\mathbb{Z}$ are base-point-independent iff the rule is affine as a function* — has been verified exhaustively for $q \leq 6$ in one neighbour-coordinate and $q \leq 3$ in two, and on 2.5×10^5 random rules at larger (q, m), with zero counterexamples; together with the telescoping proof (constant single-coordinate differences force $F(x) = F(0) + \sum_i c_i x_i$, well-defined on the torus since $qc_i \equiv 0$) this is a complete elementary result, not a genericity heuristic. The question of whether the *q-size* gauge freedom of [SM §2.7] *entails* amplitude-scale gauge resolves in the negative, and sharply: the weak reading of §2.7 (size-independence of predictions) does not reach rescaling at fixed q; the strong reading (arbitrary relabellings of $\mathbb{Z}/q\mathbb{Z}$ are gauge) *overshoots* — a nonlinear permutation conjugating a linear rule preserves cycle structure (hence emergent transition probabilities up to relabelling) while being nonlinear, so it would render a nonlinear rule gauge-equivalent to a linear one and thereby *destroy* linearity-necessity rather than establish it (verified explicitly on $\mathbb{Z}/5\mathbb{Z}$). The principled object that delivers exactly what is needed without overshooting is the **additive-automorphism** group $x \mapsto \lambda x + c$ (λ a unit) — a proper subgroup of all permutations that preserves the alphabet's "+", hence the dispersion. The upshot, in the framework's three-tier classification ([SM §8.3]): linearity is **not** promoted to a forced (Tier-1) consequence; it is a Tier-3 *definitional stipulation*, but a sharpened one — the bare assumption "the dynamics is linear" is replaced by the more primitive and better-motivated assumption "the alphabet's additive automorphisms are gauge," from which linearity follows as a certified theorem. This is the characteristic resolution shape for the

framework's foundational gaps: pressing for a proof does not close the gap to a derivation but relocates the irreducible assumption to a more primitive and more defensible place, and names it exactly.

Classification scope of the transformations. Theorem 24 establishes hidden-only coherent-channel invariance and finite or trace-class decoupled enlargement, and identifies a broader population-only hidden-fibre relabelling. Pair C refutes exhaustiveness of the *restricted coherent-channel* moves; the all-partitions classification remains open. Equality of all time-resolved visible channels does not determine the bijection up to hidden unitary, even with trivial hidden commutants.

Proposition 24.2 (Fixed-carrier block-word rigidity). *Let V, H be finite index sets and U, U' be matrices on $\mathbb{C}^V \otimes \mathbb{C}^H$. Write $B_{vw}(U)$ for their visible blocks on $\mathbb{C}^H$, and let $\mathcal{W}(U)$ be the linear span of all words in the blocks and their adjoints, including the identity. Suppose $\mathcal{W}(U) = M_{|H|}(\mathbb{C})$ and the unnormalized trace of every corresponding block-word agrees for U and U'. Then a hidden unitary W satisfies $U' = (\mathbb{1}_V \otimes W)U(\mathbb{1}_V \otimes W)^\dagger$. Conversely, hidden-unitary conjugation preserves every block-word trace, and the hypotheses imply $\mathcal{W}(U') = M_{|H|}(\mathbb{C})$.* No unitarity assumption on U, U' is needed, but the two systems must have the same hidden index set.

Proof. Equality of all word traces yields a trace-preserving unital $*$-isomorphism of the generated word spans. When the first span is the full matrix algebra, the matrix-unit construction implements it by a unitary. Necessity follows by trace invariance. The Lean witnesses for the two directions are WT2gen_oneSided and trace_wordEval_hiddenConjugate, combined in WT3_spanning_iff; blockSpanning_of_wordTracesEqual establishes spanning for the target (OIBridge/WordTraceSufficiency.lean and OIBridge/SemigroupTransfer.lean). □

Limits of the channel-family data. In the Lean-checked ST3_pairC counterexample, two permutation unitaries on six states ($|V| = 2$, $|H| = 3$) induce the same visible channel Φ_t at every time, with nontrivial visible statistics and trivial hidden commutants. Their traces are 2 and 0, so no unitary conjugates them. ST4_pairC also excludes the tested same-size decoupled-enlargement and hidden-relabelling alternative. They have unequal block-word traces (pairC_wordTraces), so they do not meet Proposition 24.2's stronger premises. These counterexamples refute the channel-family-to-hidden-conjugation implication and, under the restricted fixed-interface moves, generator exhaustiveness. They do not classify unrestricted population or all-partitions equivalence.

Open scope questions. Applying Proposition 24.2 to reconstruction requires the selected observables to determine all block-word traces and the relevant dynamics to have a full block span. A hidden unitary also need not be a finite-state relabelling of form (i). The fixed-interface coherent-channel generator proposal is refuted by Pair C under the restricted moves above; classification of population equivalence and equivalence over all admissible partitions remains

open. Whether additional coherent or interleaved multi-time data are physically available is an independent question.

4.1 Conditional reconstruction from complete word data

Multi-time records can recover more than separate one-time channels, but their observational carrier must be specified. The following standard linear-realization result [5, 6] and its native deterministic specialization recover predictive structure. They complement Proposition 24.2, whose input is coherent block-word traces, rather than imply its hypotheses.

Proposition 24.3 (Minimal linear word realizations). *Let a finite alphabet $\mathcal{A}$ have two finite-dimensional realizations $(E, a, b, (M_c))$ and $(E', a', b', (M'_c))$ over a field, with chronological convention $M_{c_1 \cdots c_k} = M_{c_k} \cdots M_{c_1}$ and $M_\epsilon = I$. Assume each is reachable,* $\operatorname{span}\{M_w a\} = E$, *and observable,* $\bigcap_u \ker(bM_u) = \{0\}$, *and similarly for the primed model. Equality $bM_w a = b' M'_w a'$ for every word implies a unique linear isomorphism S with $Sa = a'$, $SM_c = M'_c S$, and $b'S = b$. For arbitrary competing dimensions $n, m \geq 1$, equality through word length $n + m - 1$ already implies equality for all words. A nonzero series of known minimal dimension d has a scalar Hankel reconstruction using prefix and suffix lengths at most $d - 1$ and word statistics through $2d - 1$.*

Proof. Define

$$S\left(\sum_i t_i M_{w_i} a\right) = \sum_i t_i M'_{w_i} a'.$$

If the source sum is zero, every target continuation evaluates the proposed image to $\sum_i t_i b' M'_u M'_{w_i} a' = \sum_i t_i b M_u M_{w_i} a = 0$; target observability makes the image zero. Thus the map is well-defined. Reversing the construction gives its inverse by reachability. The empty word gives $Sa = a'$, appending a letter gives $SM_c = M'_c S$, and equality of evaluations gives $b'S = b$. Any such map agrees on the reachable generators, proving uniqueness.

For the length bound, the reachable spaces $R_k = \operatorname{span}\{M_w a : |w| \leq k\}$ satisfy $R_{k+1} = R_k + \sum_c M_c R_k$. Equality at two consecutive stages makes the space invariant and the sequence stable; otherwise its dimension increases. In dimension N it therefore stabilizes by $N - 1$ for nonzero a, and the zero-vector case is immediate. Apply this to the $(n + m)$-dimensional discrepancy realization with operators $M_c \oplus M'_c$, vector (a, a'), and evaluation $(b, -b')$.

In a minimal d-dimensional model choose reachable-basis prefixes w_j and observable-basis suffixes u_i, each of length at most $d - 1$ by the same filtration argument. Put $R = [M_{w_j} a]_j$ and $O = [bM_{u_i}]_i$. Then

$$H_{ij} = s(w_j u_i) = (OR)_{ij}, \qquad (H_c)_{ij} = s(w_j c u_i) = (OM_c R)_{ij}.$$

Both O and R are invertible. The reconstructed coordinates are

$$\widehat{M}_c = H^{-1} H_c, \qquad \hat{a} = H^{-1}(s(u_i))_i, \qquad \hat{b} = (s(w_j))_j.$$

These equal $R^{-1}M_cR$, $R^{-1}a$, and bR. The largest required length is $(d-1)+1+(d-1)=2d-1$, and the invertible minor proves that full Hankel rank is d. □

The alphabet must be stationary and freely composable, with every required prefix, suffix and extension accessible; an outcome-labelled instrument uses an unnormalized linear update. A restricted protocol grammar or changing apparatus requires its own state augmentation or admissible-language theorem. The dimension/rank bound is an independent hypothesis: a finite observed Hankel rank alone does not certify it. Similarity is only a linear conclusion. Even strictly positive normalized minimal stochastic models can have a unique non-monomial similarity with an inverse having negative entries. An isomorphism mapping the entire normalized classical simplex onto itself would instead permute its extreme points, but generic linear reachability and observability do not establish that stronger condition.

Proposition 24.4 (Passive finite-permutation predictive quotient). *Let P be a permutation of a nonempty finite set Ω, with deterministic passive readout $r : \Omega \to \mathcal{A}$ and initial law μ. Identify ω, η exactly when $r(P^t\omega) = r(P^t\eta)$ for every $t \geq 0$, and let Q be the resulting future-trace quotient. It inherits a permutation $\bar{P}$, readout $\bar{r}$, and prior $\bar{\mu}$. For each model being compared, assume its quotient prior $\bar{\mu}$ has full support, and let B be a known upper bound on both quotient-state counts. For $B \geq 2$, equality of all passive word probabilities through length B, counting the time-zero readout, determines $(Q, \bar{P}, \bar{r}, \bar{\mu})$ up to a unique readout-preserving conjugacy transporting the prior. For $B = 1$, one-letter data suffice. No invariance or uniformity of μ is required. The quotient's linear Hankel rank equals $|Q|$; common-length retained-word vectors and block effects generate its full nonnegative state cone and all coordinate indicators, respectively. Minimal-realization similarity between two such quotients is a permutation.*

Proof. Future-trace equivalence is forward invariant. Since P has a finite order, every negative power is also a nonnegative power, so the equivalence is backward invariant too and $\bar{P}$ is a permutation. On an N-state deterministic labelled system let C_k be the partition by its first k readouts, with C_0 indiscrete. If $C_k = C_{k+1}$, equality of labels and successor classes makes this equivalence forward invariant and no later refinement occurs. Otherwise the number of classes increases. Thus C_{N-1} is the full future partition for $N \geq 2$.

Put $\ell = B - 1 \geq 1$. Distinct quotient states are distinguished by their length-ℓ future blocks. Full support makes the common positive-probability length-ℓ words exactly the state labels W. Each $w \in W$ has exactly one symbol x with $s(wx) > 0$; its successor label is the last ℓ symbols of wx. The first symbol of w gives its readout and $s(w)$ gives its initial mass. Hence length-B data determine the labelled quotient permutation and prior. Any readout-preserving intertwiner preserves the entire future trace, so this conjugacy is unique. For $B = 1$ the sole state has identity dynamics and its one-letter law determines its label.

For the cone and rank statements choose one common separating length $\ell \geq \max(1, |Q|-1)$. In quotient coordinates set $M_x = \bar{P} D_x$, where D_x filters readout x, $a = \bar{\mu}$, and $b = \mathbf{1}^T$. These reproduce passive cylinder probabilities because the last advance cancels under $\mathbf{1}^T \bar{P} = \mathbf{1}^T$. If w_q is the length-ℓ block of state q, then

$$bM_{w_q} = e_q^T, \qquad M_{w_q} \bar{\mu} = \bar{\mu}(q) e_{\bar{P}^\ell q}.$$

Full support gives a positive coefficient, and the common power $\bar{P}^\ell$ bijectively covers all end states. Thus the retained-word rays and block effects cover the coordinate rays and indicators. The model is reachable and observable, so its Hankel rank is $|Q|$. Matching blocks and their probabilities makes the similarity of Proposition 24.3 carry the corresponding coordinate rays with unit scale, hence be a permutation. □

Access and remaining gauge gap. This is a predictive-quotient result for passive, nondisturbing records from one fixed finite reversible carrier. It refines [Main §3.4]'s existing finite-horizon canonical tail quotient; it does not infer an all-time carrier from arbitrary finite-horizon laws. Full support on the microscopic carrier is sufficient, but one rooted visible preparation $\delta_v \otimes \mu_H$ is generally not such a prior. A consistent positive mixture of every root would supply it only if that preparation family is actually accessible with the same dynamics and hidden prior. Algebraic saturation on Q does not assert externally selectable microscopic states or separate states in the same future-trace fibre. Without a known state bound, full-support uniform laws on binary de Bruijn cycles of orders k and $k+1$ agree through length k and differ later. Invisible fibres can also have different microscopic cycle structures. Neither proposition therefore supplies hidden-factor conjugacy, coherent block-word traces, a common quantum instrument theory, or completeness over all admissible partitions.

The self-contained derivation and bounded exact rational controls are retained in verification/audits/manuscript/multitime-predictive-recovery-20261028/. The native checks cover 768 four-state cases and two one-state cases, including nonstationary full-support priors, constant readouts, common-length cone/Hankel checks, and countercontrols for missing hypotheses. These are executable evidence supplementing the written proofs; neither new proposition is claimed Lean-proved.

Proposition 24.5 (Passive record fibres and state identification). *Let P be a known permutation of a nonempty finite carrier Ω, let $r : \Omega \to \mathcal{A}$ be a deterministic readout, and let μ be a known initial law with support A. Suppose the observer may retain the exact passive readings $X_j = r(P^j S_0)$. For a positive-probability word $w = (w_0, \dots, w_{k-1})$ of length $k \geq 1$, define*

$$F_w = \{s \in A : r(P^j s) = w_j \text{ for } 0 \leq j < k\}.$$

(i) A quantity $q(S_0)$ is identified by that record exactly when q is constant on F_w. In particular the initial state is identified exactly when $|F_w| = 1$. The current state $P^{k-1} S_0$ has the same singleton criterion. When $\Omega = \mathcal{C}_V \times \mathcal{C}_H$ and $r = \pi_V$,

the current hidden state is identified exactly under that same criterion. (ii) Put $N = |\Omega|$ *and* $L = \max(1, N-1)$*. Passive words of length* L *already induce the full future-trace equivalence on* Ω*, hence on* A *by restriction. Every positive-probability finite record remains compatible with at least two initial states if and only if every nonempty intersection of a full future-trace class with* A *has at least two elements.*

Proof. Conditioning gives

$$\Pr(S_0 = s \mid w) = \frac{\mu(s)\mathbf{1}_{F_w}(s)}{\sum_{u \in F_w} \mu(u)}.$$

Every point of F_w has positive mass, so a quantity is known exactly when it is constant there. Invertibility maps F_w bijectively to the current compatible-state set. For a product carrier with current visible value recorded, distinct current compatible states have distinct hidden coordinates. This proves (i).

For (ii), partition Ω by its first j readings, starting with the indiscrete partition at $j = 0$. Each next reading refines it. If two consecutive partitions agree, the classes have identical labels and successors in the same class; no later refinement occurs. Otherwise the class count strictly increases. Starting with one class, there can be at most $N-1$ strict increases, so length $N-1$ suffices for $N \geq 2$; one reading suffices when $N = 1$. Restriction to A gives the assertion without assuming that A is invariant. Each nonempty finite-record fibre is a union of restricted full future-trace classes. Thus classes of size at least two make every such fibre non-singleton; conversely a singleton full class yields a singleton length-L fibre. □

Memory and opacity are different. Under the uniform law in the swap model, $H_1 = X_0$, $X_2 = X_0$ and $I(X_0; X_2 \mid X_1) = 1$, but $H(H_1 \mid X_0, X_1) = 0$. More generally, cyclically shifting $m+1$ independent uniform bits and reading the first gives a hidden store of 2^m states, all identified by $m+1$ retained readings. At time m, $I(X_{<m}; X_{>m} \mid X_m) = m$ bits, saturating the hidden capacity, while $H(H_m \mid X_{\leq m}) = 0$. Indeed, conditional on the current bit, the preceding m readings specify all other bits, and the next m readings repeat them. C4's information is predictive memory relative to the present readout, not a lower bound on residual uncertainty after the full history.

Access and resource scope. The proposition assumes the dynamics, preparation support and retained records are available. It supplies no physical record-storage or intervention mechanism. A stored record with only R possible values cannot distinguish all $|A| > R$ admissible initial states: some fibre has at least $\lceil |A|/R \rceil$ members, although other fibres may be singletons. This pigeonhole limit is weaker than claiming ambiguity for every history. A decoupled spectator gives genuine non-singleton passive classes; fixing that spectator by preparation, reading it, or allowing a separating intervention can remove the ambiguity. Proposition 24.4 instead reconstructs an unknown predictive quotient from exact word laws under its own support and state-bound

hypotheses. Neither result identifies the physical OI model, coherent instruments or the full gauge relation. The bounded exact controls are retained in verification/audits/manuscript/hidden-state-identifiability-20261028/.

Other structural cases. The q-insensitive conclusions of [SM §2.7] are unconditional within their stated domain, while equality of transition probabilities across q remains an additional hypothesis. Theorems 1–6 retain their conditional structural graph conclusions without implying equality of full channels; Standard Model representation content requires octahedral structure. In the selected wave-rule branch a layer swap T satisfying $\varphi^{-1} = T \circ \varphi \circ T^{-1}$ supplies a conditional time-reversal relation, **not** a hidden-only instance of generator (i). A global phase of the emergent unitary is not a separate finite-bijection operation.

A proposed gauge hierarchy, with an open substratum step. The three descriptive levels below carry distinct, stated equivalence notions. Their nesting as groups and a complete projection of substratum equivalences to emergent symmetries require additional hypotheses; neither follows from Theorem 24 or Proposition 24.2 alone.

Level 3 (substratum): At a fixed visible/hidden interface, Theorem 24(i),(iii) exhibits exact coherent-channel invariances under a uniform hidden-factor relabelling and any well-defined decoupled normalized ancillary enlargement, and population invariance under fibre-dependent hidden relabellings. Its alphabet and graph clauses concern narrower structural comparisons, and a complete all-partitions equivalence relation has not been characterized. The notation $\mathcal{G}_{\mathrm{sub}}$ denotes the proposed full substratum observational gauge structure only once its observables, relabellings and partition correspondence have been specified; it is not established here as generated by (i)–(iv).

Level 2 (emergent QFT): The selected cubic branch supplies an exact $(3, 2, 1)$ link representation and a conditional $U(3) \times U(2) \times U(1)$ condensate stabilizer. The SM Lie algebra and faithful charged-family image require the specified correlated quotient and matter action [SM §4.4–§4.9]. A complete map from substratum observational equivalence to that physical gauge action remains open.

Level 1 (emergent Hamiltonian): The diagonal-unitary transformation $H \mapsto DHD^{\dagger}$ preserves the transition probabilities in the specified measurement basis. Energy shifts and the antiunitary reflection also preserve that data; completeness of the stated alternatives is conditional on H-BG and the gap/overlap hypotheses [GR §3.3].

These are compatible descriptive levels, not a proved inclusion chain G1 $\subseteq$ G2 $\subseteq$ G3. Establishing such inclusions requires definitions and maps that preserve the same observational data at each step, rather than only graph invariants or fixed-interface channels. The substratum equivalence, unlike an emergent Lagrangian gauge symmetry, concerns representations of a finite bijection and specified observer access; no substratum Lagrangian is postulated.

5. Synthesis: Three Projections of (S, φ)

Theorem 23 establishes the stated local lattice/gauge outputs **on the selected cubic branch** under its physical and mathematical assumptions; uniqueness of that branch modulo an observational gauge group is not established. M1-B and H-Bell separately condition the existence of a measurement-independent Bell-violating completion; Bell-inclusive uniqueness is open. The fixed-interface results of §4 supply examples of observable-preserving transformations; a general equivalence classification and a complete projection to Standard Model gauge symmetry are not established. This section makes the synthesis claim explicit: quantum mechanics, general relativity, and the arrow of time are not three independent theories but three projections of the same finite deterministic construction.

5.1 The three projections

The framework's derivations apply the same trace-out machinery to the same (S, φ) along three projections: a visible-sector projection that produces the reduced description admitting the quantum representation ([Main]) and the Standard Model ([SM]); a boundary-thermodynamic projection that produces general relativity ([GR]); and an arrow-of-time projection (§5.4) that produces the emergent cosmological, thermodynamic, memory, and measurement arrows from a single structural clock and the C1–C4 observer-selection theorem of [Main, §4.6].

Visible-sector projection. At the level of the embedded observer's epistemic access, the trace-out over the hidden sector produces the exact finite observable-law correspondence established in [Main]: $S \iff D \iff Q_{\mathrm{fb}}$. The fixed-basis unitary/Born representation is therefore exact and bidirectional but universal. The nontrivial OI content is history readback: C4 forces conditional memory and hidden predictive structure in every faithful completion; C1 and C3 diagnose the mediation and its capacity, while C2-structural, the persistence of the record to readback, holds by reversibility in every bijective realization and constrains coarse-grained descriptions of the hidden sector. In the spatial theory the dependency graph additionally supplies exact causal cones for finite-range couplings. Full equality with the standard coherent instrument/composite theory remains the narrower operational-lifting statement of [Main §3.4]. Bell is separate: the measurement-independent deterministic completion reaches quantum Bell violation only by ontic parameter dependence, with operational no-signaling [Main §3.3]. Applied to a cubic lattice with the wave equation as substratum dynamics — itself uniquely selected among second-order reversible nearest-neighbor dynamics by center independence, isotropy, and linearity ([SM, §4.1]) — this projection produces the Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ (as the commutant of the coupling matrix, [SM, §§4.4–4.6]), three generations (physical reading conditional on H-spin', [SM §4.7]; chirality on H-χ', [SM §4.8]) and the

Higgs as a composite scalar in the singlet staggered taste ([SM, §4.7], Theorems 8–11), and anomaly-free hypercharges ([SM, §4.9], Theorems 14–15). Twenty-two quantitative predictions follow ([SM, §7]), including the Cabibbo angle from a single Brillouin-zone distance, the Koide angle from a cubic-group quadratic Casimir, all three PMNS angles within 1.1σ (vs NuFIT 6.0), six fermion masses from one empirical input to better than 1%, and all three SM gauge couplings at M_Z ([SM, §6.3]).

Boundary-thermodynamic projection. At the level of the partition boundary, classical horizon thermodynamics — the temperature, entropy, and dynamical evolution of the causal horizon as a gravitating object — produces general relativity through Jacobson's thermodynamic argument, conditionally on the open covariant step ([GR, §8.7]): the Clausius relation $dE = T\,dS$ applied at local causal horizons yields Einstein's equations with the same structural inputs that give the visible-sector projection its form. Applied to the cosmological horizon as a causal partition ([GR, §2]), this projection determines the emergent action scale $\hbar = c^3\epsilon^2/(4G\sigma)$ from thermal self-consistency ([GR, §3]), fixes the discreteness scale $\epsilon = 2\sqrt{\sigma}\,l_p$, for an entropy σ per boundary mode ($\epsilon = 2l_p$ at one nat per mode), as the unique simultaneous solution ([GR, §4]), recovers the Bekenstein-Hawking entropy $S = A/(4l_p^2)$, the mode count reproducing it once the calibration fixes $\epsilon/\sqrt{\sigma} = 2l_p$ and the 1/4 being the ratio $2\pi/8\pi$ of the KMS and first-law normalizations ([GR, §5]) — the action scale, the discreteness scale and the coefficient conditional on H-slope with the horizon and frame conditions of [GR, §2, §8.5]; GW250114 confirms the classical area theorem, the coefficient itself untested directly — and addresses the cosmological constant problem by identifying the quantum vacuum energy and the effective Λ as properties of logically distinct levels of description rather than commensurable quantities whose 10^{122} ratio requires cancellation ([GR, §6]), the descriptive distinction established and the dissolution conditional on the open coupling step ([GR, §6.2]). The framework carries dark energy in running-vacuum form — the functional form the local expansion shared by every smooth model, the content the Type II channel with magnitude $|\nu| \sim 10^{-32}$ (not derived at theorem level) — consistent with the Bertini et al. (2025) direct DESI-DR2-era fit at 2σ (which is consistent with the ΛCDM limit); a ~95% dark sector corollary as a structural feature rather than a new particle species; and a MOND acceleration scale $a_0 = cH/6$ from entropy displacement at the boundary ([GR, §7]).

The proposed projections take the same hypothesized substratum (S, φ) and observer partition as starting inputs. Their distinct constructions share partial-observation machinery, but neither the complete substratum gauge group nor all gravitational coupling steps are derived from those inputs. They differ only in which structure of the embedded observer's description they derive: the visible-sector projection (§5.1, §5.3) works on the bulk dynamics and produces the quantum-representable reduced description and the Standard Model; the boundary-thermodynamic projection (§5.1) works on the classical thermal data at the partition boundary and produces general relativity; the arrow-of-time projection (§5.4) works on the horizon-clock parameter along observer worldlines

and produces the cascade of emergent arrows.

The projections and the Poincaré structure of the partition. The three projections are organized by how the observer-worldline-centered partition treats the Poincaré group, which makes the boost/rotation result of [Main §3.5] one facet of a larger pattern. The partition selects a preferred event — a spatial origin and a rest frame, jointly the observer's worldline — so the compact rotations about that worldline are not excluded by the partition's rest-frame data, while the non-compact generators are broken, given that the coarse-graining map determines the horizon foliation ([Main §3.5]); spatial isotropy enters as an empirical input. Boosts, tied to the selected rest frame, surface as the cosmic rest frame (operationally the CMB frame) and the boost-sector Lorentz residual ([SM §3.1]); spatial translations, which merely relabel the observer's origin, are broken only observer-relatively and are recovered as spatial homogeneity — the equivalence of all worldline-centered descriptions; and time translation is broken intrinsically by the growth of the horizon, which is the boundary-thermodynamic and arrow projections (§5.4, [GR]). The partition's symmetry-breaking signature — isotropy not broken by the partition, a preferred rest frame, spatial homogeneity, a distinguished cosmic time — is the symmetry structure of a Friedmann-Robertson-Walker cosmology. The framework does not derive that structure; it exhibits the partition's symmetry-breaking pattern as compatible with it, with ordinary momentum conservation holding in the visible sector up to corrections of order (length / horizon radius).

5.2 The QM-GR incompatibility as a category error

The conventional formulation of the QM-GR unification problem treats quantum mechanics and general relativity as two competing theories in the same logical category — two attempts to describe the same regime — and asks how to merge them into a single mathematically consistent framework. Every program of unification proceeds from this assumption: quantum gravity programs treat the metric as a quantum field to be quantized; geometric programs treat the wave function as a structure on the spacetime manifold; emergent programs treat one of the two as derived from a deeper substrate that recovers the other. The persistent failure of all three approaches to produce a quantitatively confirmed unification suggests that the underlying assumption is wrong.

In the present framework, the assumption is wrong because quantum mechanics and general relativity occupy different positions in the trace-out hierarchy. They are not two theories of the same regime at all — they are two projections of the same construction, viewed at different levels. The visible-sector projection that produces the quantum-representable reduced description and the boundary-thermodynamic projection that produces GR are not in tension because they refer to *different objects in the construction*. The wave function is a property of the embedded observer's compressed description of the visible sector; the metric is a property of the classical dynamics of the partition boundary. These are not two descriptions of the same physical system at different levels of approximation.

They are descriptions of two different substructures of the same total object (S, φ).

The analogy that captures the structure most clearly is the one between a thermodynamic and a statistical-mechanical description of a gas. The thermodynamic description deals with pressure, temperature, and entropy; the statistical-mechanical description deals with particle positions, momenta, and microstate counting. These are not two competing theories of the gas, and the question "how do we unify thermodynamics and statistical mechanics?" is not a coherent question — they describe different things about the same system, and the relationship between them is one of projection (statistical mechanics produces thermodynamic quantities by averaging) rather than unification. The QM-GR relationship in the present framework is structurally similar: GR describes the thermodynamic (boundary, classical, deterministic) projection of (S, φ), and QM describes the statistical (visible-sector, compressed, stochastic-emergent) projection. The two are related by a definite procedure (the trace-out under conditions C1–C4), and the question "how do we unify them?" is dissolved rather than answered.

This is not the same as saying that quantum mechanics is "more fundamental" than general relativity, or vice versa. Both are emergent. Both depend on the trace-out and the partition. Both are derived rather than fundamental. The fundamental object is (S, φ) — a finite set with a deterministic bijection — and neither QM nor GR is a feature of (S, φ) itself. They are features of how an embedded observer or a horizon-bounded thermodynamic regime sees (S, φ) from inside. (Nor is the ranking question recoverable from inside: by the factoring argument ([GR §6.3]), no criterion grounded in the accessible observations can separate two descriptions faithful to the same data — the fundamentality question is answered at the level of (S, φ), not adjudicated between its projections.)

5.3 The three-level hierarchy made explicit

The restricted observational invariances of §4 motivate a relationship between the visible and boundary projections; they do not establish a complete gauge-group image under trace-out. At the substratum level the chosen object is a bijection together with an observer partition, and the proposed projection connects it to two separately described structures: the emergent quantum field theory at the visible-sector level, with the Standard Model gauge group as the commutant of the coupling matrix ([SM, §4.4], Theorem 5) and the Standard Model representations as the cubic decomposition of the link directions ([SM, §4.6], Theorem 7); and the classical horizon thermodynamics at the boundary level, with the metric, the surface gravity, the entropy, and the temperature determined by the partition geometry and the area-law theorem ([GR, §3]). Both derived structures inherit the framework's structural inputs from the substratum, and both are exact rather than approximate at the lattice level.

The three-level structure is a proposed relationship between distinct equivalence

notions, with only the explicitly stated parts established. Fixed-interface hidden relabellings and finite exact decoupled extensions are visible-channel invariances (§4); the Standard Model commutant is obtained from the cubic representation construction; and the emergent Hamiltonian's D-gauge preserves transition data. A full group inclusion and projection between these notions remain open, as does the H-BG-dependent completeness of the Hamiltonian classification [GR §3.3].

The boundary-thermodynamic projection that produces GR is a parallel structure at Level 2 — but acting on the *partition* rather than on the internal field content. Where the emergent QFT is the trace-out's effect on the bulk dynamics, the classical horizon thermodynamics is the trace-out's effect on the partition geometry itself. Both are at Level 2. Both descend from Level 3. The framework's claim is that the SM gauge group and Einstein's equations are not two independent theoretical inputs but two co-derived structures, both consequences of the same underlying (S, φ) and the same trace-out under the same conditions C1–C4; on the gravitational side the covariant field equations are the open Level-G3 step ([GR, §8.7]).

5.4 The arrow of time as a third projection

After Main §4.6 (structural observer-selection theorem), the arrow of time joins quantum mechanics and general relativity as a third emergent structure that the framework derives from the same underlying (S, φ) and the same trace-out machinery. Before §4.6, the arrow of time was treated as external to the derivation chain — a past hypothesis or an anthropic selection, imposed on the framework rather than produced by it. With §4.6 in place, the observer's confinement to the non-equilibrium phase of (S, φ) is a structural consequence of the effective-mixing hypothesis, for sufficiently fast-mixing equilibria ($\tau_{\mathrm{mix}} \ll \tau_S$), in the coarse-grained description generated by iterating the one-step effective channel, and every arrow the observer sees descends from this confinement together with the horizon growth that the accelerating phase produces. The third projection has the same form as the first two: a feature of how the embedded observer sees (S, φ) from inside, rather than a feature of (S, φ) itself.

A distinction should be kept in view here. The *existence* of the temporal domain — that differentiation recurs, that there is more than one moment at all — is not a projection of (S, φ) but the framework's second foundational axiom (the recurrence axiom of the two-axiom observation base; see [Main §1.2] and Part I of the companion paper *Physics Modulo Gauge*). What is a projection, and what this section derives, is the *arrow* — the directedness of time within that domain. The framework derives the arrow; it posits the existence of the domain. Conflating the two would overstate what the projection delivers: §5.4 produces directedness, not the bare existence of succession.

Horizon growth as the structural clock. The framework's primary arrow is the cosmological horizon entropy $S_{\mathrm{dS}}(t) = A(t)/(4l_p^2)$, monotonically increasing along any observer worldline in the Λ-dominated phase. The monotonicity is a

feature of the causal partition rather than of any local frame: every observer in the accelerating phase agrees that S_{dS} increases, just as every observer in a Schwarzschild geometry agrees on the exterior direction. The horizon clock is observer-independent, continuous, and monotonic for as long as $\ddot{a} > 0$, which the framework identifies as the regime of interest for the derivations of [GR]. At the substrate level, (S, φ) has no preferred direction: φ and φ^{-1} are equally valid as forward-time maps. At the observer level, the horizon clock picks out a direction, and all other arrows align with it.

The layered cascade. Four derived arrows align with the horizon clock through a structural cascade. *Cosmological:* $S_{\mathrm{dS}}(t)$ monotonic in the Λ-dominated phase, as above. *Thermodynamic:* via Jacobson's coupling $\rho_s = \rho_{\mathrm{crit}}$ ([GR, §3]), the entropy of any horizon-bounded system increases with the observer's clock — the thermodynamic arrow descends from the structural arrow without an independent postulate. *Memory:* C2 (memory persistence) implies the observer's hidden sector preserves correlations forward along the clock rather than backward, because the coarse-grained mixing time of the hidden-sector dynamics is uniform in the two directions but the observer's measurement protocol commits to one. *Measurement:* the observer records "past" outcomes and prepares "future" states — an asymmetry inherited from the memory layer, not added on top of it. The chain grounds in one structural fact (horizon growth) and one structural constraint (C1–C4 observer selection via Main §4.6, conditional on its effective-mixing hypothesis and stated for the coarse-grained description generated by iterating the one-step effective channel). The effective-mixing hypothesis is the premise that takes the place of an initial-condition axiom.

What the framework derives, and the directional-language subtlety. The framework's quantitative content at the level of Main §4.6 is *symmetric* in time direction — the spectral-gap bound $\tau_B(V) \leq \tau_{\mathrm{mix}}(\Sigma_{\mathrm{loc}}(V)) + O(\epsilon)$, stated for the coarse-grained description generated by iterating the one-step effective channel, applies equally in both directions along any observer worldline. What is *imposed* by the observer, not derived, is the convention of which direction along the horizon-clock parameter to call "future." The two directions are gauge-equivalent at the substrate level ($\varphi \leftrightarrow \varphi^{-1}$), and the observer's choice of convention is the residual freedom after all physical content has been extracted. The framework therefore derives the existence of a monotonic parameter along observer worldlines, not a substrate-level arrow of time. This distinction answers Ellis's objection (a bijective substratum cannot carry an arrow of time) by agreeing with the objection's premise while deriving what the observer actually sees from the structure the substratum does have.

Scope note: causal topology as load-bearing. The memory-arrow step uses "the observer's worldline has a causal topology" as the carrier of the asymmetry. This is not a new axiom — [Main, §1.3] already assumes a causal topology on the partition — but it is load-bearing for the cascade and is noted explicitly here rather than treated as obvious. A framework in which the ob-

server's worldline had no causal topology could not run the memory-arrow step, and the cascade would halt at the thermodynamic layer.

Scope note: the arrow derivation is conditional on the ordering of moments. The cascade derives a *direction* along the observer's worldline. It presupposes that the worldline's moments are *ordered* — that they form a sequence — since a monotonic parameter is monotonic with respect to an ordering. This ordering is distinct from the bare existence of more than one moment: the framework's foundational treatment (Part I of the companion paper *Physics Modulo Gauge*) axiomatizes the existence of a temporal domain but treats the *ordering* of its moments as a separate question, which it leaves open. The arrow derivation of this section is therefore conditional: it derives the direction *given* an ordering, and inherits the open status of the ordering question. What this section establishes is the conditional — ordered moments yield a derived monotonic parameter and an observer-imposed direction-convention — not an unconditional derivation of temporal directedness from the substratum alone. The causal topology of the worldline reduces to this ordering together with the dynamics φ (it introduces no commitment beyond the ordering), so the arrow's single open dependency is the ordering question and nothing further.

CP violation as the residual Ellis-class objection. The layered cascade closes the Ellis-axis objection for the arrow of time but does not automatically close the related question of CP violation. The substratum is T-symmetric (load-bearing for the reciprocity of visible transition counts, [SM, §5]; it does not by itself fix $\bar{\theta}$), and observed CP violation in CKM and kaon/B systems is a flavor-basis feature of the specific bijection rather than a consequence of the cascade. The framework treats flavor-sector CP phases as solution-specific inputs — see [SM, §5.4] for the scope statement and §6.4 "what does not falsify" for the commitment that the framework makes no prediction of δ_{CKM}, δ_{PMNS}, or the Wolfenstein (ρ, η).

5.5 What the synthesis does and does not claim

The synthesis claim is structural and architectural rather than empirical. It does not claim that the framework predicts gravitational phenomena that conventional QFT plus GR cannot reproduce — most of [GR]'s predictions (the BH area law, the basic properties of horizon thermodynamics, the broad shape of dark energy phenomenology) are also recovered by other approaches. The framework's empirical content is concentrated in [SM] (the SM derivation, twenty-two quantitative predictions) and in the specific GR-side predictions of [GR] (the Type II running-vacuum channel with $|\nu| \sim 10^{-32}$, MOND $a_0 = cH/6$ with its dimensional factor imported, dark sector concordance), which collectively distinguish the framework from standard ΛCDM plus the Standard Model.

The three constructions share a candidate model and partial-observation machinery, but use different additional premises. The cubic gauge calculation in [SM] and the horizon mode count in [GR] are not the same mathematical operation,

and their physical interpretation is conditional on their respective hypotheses. Their common-substratum interpretation is a proposed relationship between the constructions, not a proof of a unique joint physical realization.

The proposed synthesis reframes the relation between quantum representations and gravitational thermodynamics within a candidate deterministic model. It does not remove the physical compatibility and reconstruction problems from the research programme. The Standard Model gauge factors are obtained in the selected cubic representation analysis under its own hypotheses. Identifying them as the complete image of a full substratum gauge group requires an observational carrier and a comparison map not established in §4. Neither that image nor uniqueness of the physical branch follows from the common use of (S, φ).

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6. Discussion

The exact finite-law representation, conditional local construction and scoped observational invariances address different questions. Theorem 23 organizes outputs of a selected branch; Theorem 24 distinguishes transformations preserving coherent channels from those preserving only populations, while all-partitions classification remains open. These results support a structural interpretation of specified model observables. They do not establish that mathematics exhausts reality, that every microscopic distinction is empirically inaccessible, or that the quantum/gravitational compatibility problem is solved.

6.1 Ontic structural realism on the equivalence class

The structural-realist proposal of §1.1 concerns the conditional local outputs of the selected model and invariants of a specified observational carrier. Theorem 23 does not uniquely identify that model among all substrata satisfying E1–E7 and A1–A6. H-Bell separately conditions a common Bell-compatible completion; it is not the only missing step in a full reconstruction. Realism about one complete microscopic bijection therefore requires physical selection and equivalence results beyond the present theorems.

This is the strongest claim the framework's theorems license at this level without overclaiming. Two alternative framings merit explicit comparison.

Specific-substratum realism is the commitment that observation singles out one microscopic bijection φ. Neither Theorem 23's conditional local-branch outputs nor Theorem 24's scoped invariances establish that commitment. For a **specified** observer interface, the proven hidden-only relabellings and exactly decoupled ancillary enlargements preserve coherent channels; fibre-dependent hidden relabellings preserve rooted populations without necessarily preserving coherences. The alphabet and graph comparisons preserve only their stated structural contents. A measurement restricted to a carrier cannot distinguish

representatives related by a transformation proved to preserve that carrier, but no complete all-partitions gauge classification is available to rule out further observable distinctions.

Pure explanatory economy — the position that the selected cubic construction provides an economical account of several observed structures without committing to one microscopic ontology — remains a defensible interpretation of the current results. Theorem 23 gives conditional lattice/gauge outputs on that selected branch, but does not prove that the empirical and structural inputs uniquely identify it modulo a complete gauge group. The distinction between what the construction explains within its assumptions and whether nature selects that branch is substantive, rather than a question settled by observational redundancy.

The wave function, Hilbert space, gauge fields and metric enter particular descriptions of the candidate model, with their own observer and physical hypotheses. This motivates comparing descriptions at different levels, but does not prove that they are all projections of a uniquely reconstructed observational quotient. Whether a change of representation is physically redundant is assessed relative to the specified observable carrier and its readout map.

Two ontological readings of the same model are indistinguishable by measurements of a specified carrier if they stipulate identical preparation/readout data. This is a conditional factoring statement. Theorem 23 does not supply identical data for every candidate microscopic interpretation, and Theorem 24 does not classify all transformations preserving complete physical observations. A structural reading of Wigner's puzzle is therefore an interpretation of the construction, not an additional reconstruction theorem.

Multi-level structural realism. The structural-realism commitment is not single-level. [Structure] establishes that the framework operates at multiple levels of structural realism with different content at each level. The hierarchy is two-dimensional: an observation-hierarchy axis (vertical depth — from foundational to specific) and a gauge-hierarchy axis (horizontal breadth — from narrow to wide gauge equivalence). The combination produces a structure of realisms operating at multiple intersections.

At the broadest level — universality-class equivalence across structural classes (Level G4 in [Structure §2]) operating at the partial-trace observational features sub-class (Level C in [Structure §2]) — what is real is the algebra-channel structure of partial-trace observation: availability of a fixed-basis Born-form representation, channel-level unitarity in that representation, the non-Markovian marginal in the readback sector, and the commutant gauge-invariance pattern. These are representation/observation features shared across the stated observer-admitting class; the claim does not independently select the quadratic exponent or close the coherent local operational lift.

At intermediate levels, the interpretation concerns the Standard Model gauge algebra calculated on the selected branch and the specified algebra/channel struc-

tures. The physical gauge-group reading retains the carrier, amplitude-scale and matter hypotheses of [SM]; it does not establish a unique full substratum gauge image.

At the model-specific level (Level D × Level G3 in [Structure §2]), Theorem 23 supplies conditional outputs of the chosen cubic realization. It does not identify every compatible model with that realization modulo a complete gauge group. The classifications of Cabibbo, Koide and gravitational quantities remain those of their individual derivation chains in [SM] and [GR]. They do not establish the missing branch selector or Bell-compatible completion.

The proposed hierarchy distinguishes model-specific assumptions from structures preserved by specified observational comparisons. At the local level the mathematical outputs are conditional on the selected branch. At broader levels, identifying equivalent descriptions requires the relevant observable structures and explicit maps. A complete physical equivalence class and its ontological interpretation remain separate questions.

The class-specific phenomenology is concentrated at Level D × Level G3: the selected cubic realization, its conditional gauge/matter interpretation and its classified quantitative comparisons. Comparisons with other programmes must specify which observational carrier is shared. The hierarchy supplies a way to pose these comparisons; Theorem 23 and Theorem 24 do not establish all inter-level identifications.

6.2 The ontological hierarchy

The triple (S, φ, V) generates every concept in fundamental physics, not as independent substances but as different aspects of the same structure. **Space** is the coupling structure of φ — the graph G_φ determined by which degrees of freedom affect which others ([SM, §2.4]). **Matter** is the state — localized patterns that propagate through the coupling graph. **Energy** is the rate of change under iteration. **Time** is the iteration itself. **Quantum mechanics** is the observer's compressed description of the visible sector. **General relativity** is the thermodynamic limit of the coupling structure. **Conservation laws** are emergent: energy conservation (Noether) is what information conservation (bijectivity) looks like in the emergent quantum description. None of these are independent entities; they are descriptions of (S, φ, V) at different scales.

6.3 The measurement problem

On the structural reading, the measurement problem is dissolved. The wave function is not a component of (S, φ, V) — it is a derived object. Since it is derived, not fundamental, asking "does it collapse?" is asking about the behavior of a compression artifact. In the double-slit experiment, the particle traverses a single slit in the deterministic substratum. In Wigner's friend, the Friend has a definite outcome; Wigner's superposition reflects his epistemic deficit.

Branching is forbidden by the rigidity of φ. A fixed bijection on a finite set has exactly one trajectory from any initial state. There is no point at which the trajectory splits. The appearance of branching in the emergent quantum description reflects the observer's uncertainty about which trajectory they are on (because they cannot see the hidden sector), not a physical splitting of worlds.

Bell correlations force a sharper distinction. For any **finite-range Bell-relevant** substratum, the dependency graph gives an exact causal cone; if a response-complete deterministic completion also uses one setting-independent pre-setting ensemble, Bell factorization follows and CHSH is capped at 2. The framework therefore does not obtain quantum Bell violation from screening plus indivisibility. It keeps measurement independence and takes ontic parameter dependence: Bell-violating substrata have Bell-relevant coupling range exceeding the wings' separation, while the averaged operational statistics remain no-signaling [Main §3.3]. The stochastic causal-local indivisible Tsirelson theorem is retained only at that weaker stochastic layer.

6.4 Pre-registered falsification conditions

The framework's structural content — the equivalence class $[(S, \varphi)]/\mathcal{G}_{\text{sub}}$, the emergent Standard Model structure, and the co-derivation of quantum mechanics and general relativity — makes specific commitments whose future observational status can now be stated explicitly. This subsection pre-registers the conditions under which the framework would be falsified, separated from properties of the specific bijection φ that the framework does not claim to determine.

The distinction matters for two reasons. First, an explicitly pre-registered falsification schedule is the empirical counterpart of the methodological discipline developed in §§3–5: a framework that derives rather than posits must be willing to specify what would invalidate the derivation. Second, pre-registration separates structural commitments — which the framework stands or falls on — from solution-specific properties that lie outside its scope, and which continued openness on does not undermine the framework.

Class A: Structural theorems. The following are claims about the equivalence class itself and would falsify the framework's reconstruction if violated.

- *Strong CP.* The construction does not fix $\bar{\theta}$ (SM §5.3, Theorem 21: T-invariance leaves $\theta \in \{0, \pi\}$ and does not make the emergent Hamiltonian T-invariant), so the framework makes no commitment on the neutron electric dipole moment; reciprocity leaves the Jarlskog invariant free, and a mechanism making Y_u and Y_d simultaneously real would force $J = 0$ — the falsification control on H-det proposals (SM §5.3–5.5); a measured $\bar{\theta} \neq 0$ would constrain mechanisms proposed for H-det, not the T-symmetry of the substratum, which is compatible with any value of $\bar{\theta}$.
- *Gauge group SU(3) × SU(2) × U(1) exactly.* Any observation of a stable beyond-Standard-Model gauge structure — an additional Z' interpretable as a new gauge boson rather than a composite resonance, stable exotic mat-

ter transforming under a novel gauge group, or any confirmed extension of the visible-sector gauge content — falsifies the cubic-group decomposition of SM §4.6.

- *Three fermion generations.* Any observation of a stable fourth-generation fermion with Standard Model quantum numbers falsifies the taste-reduction derivation (SM §4.7). Non-stable exotic states consistent with composite or resonance interpretation do not.
- *Majorana neutrinos with normal ordering.* Inverted mass ordering, confirmed to high significance, falsifies the taste-breaking derivation of the neutrino mass matrix (SM §8.4). Confirmed Dirac nature of the neutrino (e.g., a null result for neutrinoless double-beta decay at the sensitivity that would require Majorana masses well below the framework's prediction) falsifies the prediction that no right-handed neutrino exists in the spectrum.
- *Bell/Tsirelson test.* For the cited **stochastic causal-local indivisible** sector, $S_{\mathrm{CHSH}} \leq 2\sqrt{2}$. At the deterministic level, measurement independence plus ontic parameter independence instead gives the Bell-local ceiling 2, so the framework's Bell-violating branch is ontically parameter-dependent [Main §3.3]. A confirmed loophole-free excess above $2\sqrt{2}$ would falsify standard quantum mechanics and any claim that the operational lift lands in the standard quantum Bell set; it would not by itself falsify P-indivisibility. Bare C1–C4 or graph locality do not enforce Tsirelson.
- *Leading Lorentz-invariance-violation coefficient.* The cubic-lattice dispersion relation ([SM, §4.4] Theorem 4, generalized to d = 3: $\cos(\omega\epsilon) = \frac{1}{3}\sum_{i=1}^{3}\cos(k_i\epsilon)$ for a massless mode) yields standard emergent dispersion $\omega = |k|$ at leading order. Series expansion gives the first correction as a direction-dependent quadratic term $\omega^2 = k^2[1 - 2\delta(\hat{k})(k\epsilon)^2 + O((k\epsilon)^4)]$, with $\delta(\hat{k}) = \frac{1}{24}\sum_i \hat{k}_i^4 - \frac{1}{72}$ evaluated on the rigid $\mathbb{Z}^3$ representative. The rigid-lattice anisotropy (values $\delta_{[100]} = 1/36$, $\delta_{[110]} = 1/144$, $\delta_{[111]} = 0$ — the dispersion is exactly linear along the body diagonals, $\omega = k/\sqrt{3}$ to all orders) is a *physical, direction-dependent* prediction. Its absolute orientation in space is gauge: generator (iv) of §4 gauges the rigid orientation of the cubic representative. The anisotropy *pattern* — its fourfold $\ell = 4$ cubic-harmonic structure and the direction-differences $\delta_{[100]} - \delta_{[111]}$ — is gauge-invariant, since generator (iv) preserves the octahedral structure and does not identify $\mathbb{Z}^3$ with a generic isotropic graph (which lacks the octahedral structure the gauge group requires, [SM §4.5–4.6]). The orientation-averaged coefficient $\langle\delta\rangle = 1/90$ is a direction-summary of this anisotropy, and the framework predicts the emergent dispersion

$$\omega^2 = k^2\left[1 - \frac{(k\epsilon)^2}{45} + O\left((k\epsilon)^4\right)\right] = k^2\left[1 - \frac{4\sigma}{45}\left(\frac{E}{M_{\mathrm{Pl}}}\right)^2 + O\left(\left(\frac{E}{M_{\mathrm{Pl}}}\right)^4\right)\right]$$

after using $\epsilon = 2\sqrt{\sigma}\, l_p$ and $k\epsilon = 2\sqrt{\sigma}\, E/M_{\mathrm{Pl}}$; the coefficient of $(E/M_{\mathrm{Pl}})^2$ is $4/45$ at $\sigma = 1$. Three structural features: (i) *subluminal*, since $\delta(\hat{k}) \geq 0$ ev-

erywhere on the sphere ($\langle\delta\rangle > 0$ is invariant under orientation-averaging); (ii) *no linear-in-E/M_{Pl} term*, because the even-parity cosines in the dispersion admit no $\mathcal{O}(k^3)$ contribution — any demonstration of linear LIV at any scale therefore falsifies the framework's substrate-level commitment to the cubic-lattice wave equation; (iii) *specific lattice coefficient* $1/45$ of $(k\epsilon)^2$, fixed by the cubic-lattice coordination and isotropy-gauge average, not tunable. Features (i)–(iii) and the anisotropy pattern are the Class A content, at tree level under the stated assumptions; the directional ratios $\delta_{[100]} : \delta_{[110]} : \delta_{[111]} = 4 : 1 : 0$ are independent of σ. The physical amplitude, the coefficient $4\sigma/45$ of $(E/M_{\rm Pl})^2$, carries the undetermined entropy σ per boundary mode and is not a parameter-free numerical prediction: a measurement of it would constrain σ rather than falsify the coefficient independently. Protection of the tree-level coefficient in the interacting theory is open ([SM §3.1]). Current quadratic-LIV bounds from GRB time-of-flight reach only $E_{\rm QG} \gtrsim 10^6$ GeV (roughly thirteen orders of magnitude below $M_{\rm Pl}$), so the amplitude $4\sigma/45$ is not yet near-term testable at the magnitude level; linear-LIV tests, by contrast, are already at sensitivities where any positive detection falsifies.

Class B: Parameter-free retrodictions. The framework produces a set of numerical predictions that follow from the structural content with no adjustable parameters beyond a small set of acknowledged empirical inputs (m_s for the overall fermion mass scale, m_t for the Higgs RGE running, η for the Jarlskog phase, $Q_{\rm down}$ for the down-sector Koide normalization). These predictions are stratified into five sub-classes reflecting their structural status:

Class B-S (structural). One prediction is a Layer 1 structural retrodiction with no open link specific to the entry; like every Class B entry it rests on the matter-sector hypotheses H-spin' and H-χ':

- The Wolfenstein parameter $A = \sqrt{2/3}$ ([SM §7.1]).

A global-fit A outside $\sqrt{2/3}\,(1\pm\lambda^2)$ would strain the leading-order identification, while the protected $\sin\theta_{13} = A^2\lambda$ comparison tests the geometry directly ([SM §7.1]).

Class B-C (conditional structural). Two predictions are Layer 1 retrodictions with no fitted input whose derivation chain has one named open link:

- The Cabibbo angle $\lambda = 1/(\pi\sqrt{2})$ ([SM §7.1]) — the chirality verification of the taste-changing vertex's spinor structure is closed via Mason et al. (HPQCD, hep-lat/0209152), establishing that taste-changing transitions in staggered quarks have spinor structure given by a combination of γ_μ and $\gamma_{5\mu}$ (both chirality-preserving); the identification leaves the emergent-theory dressing open, and the prediction is conditional on it.
- The Koide angle $\theta_0 = C_2/d^2 = 2/9$ ([SM §7.2]) — matched to observation (0.001%) and the cleanest structural derivation (cubic-group quadratic Casimir over bandwidth squared, with the §7.2 uniqueness table enumer-

ating ten alternative dimensionless ratios that fail to match); it is conditional with the Koide amplitude, whose value $A^2 = C_2$ is not yet derived from the dynamics ([SM §7.2]).

A precision-upgrade violation would falsify the identification or the closure on which the entry is conditional.

Class B-L (layered conditional, pending open derivations). Six predictions (the two PMNS angles counted separately) follow from the structural form combined with one Layer-2 substrate input, one empirical input, or one named condition on the horizon reservoir. Each has an explicit derivational gap whose closure would move the prediction to B-S:

- The mass ratio $m_d/m_s = 1/(2\pi^2)$ ([SM §7.1]), its conversion from the Cabibbo scale running through the Gatto-Sartori-Tonin relation, an external empirical relation whose status the chain inherits.
- The mass ratio $m_u/m_d = \sqrt{\theta_0} = \sqrt{2/9}$ ([SM §7.2]) pending the explicit derivation of the "different channels" mechanism producing the square root, structurally analogous to the open OI-vertex 1-loop derivation for $K = 1/2$ ([SM §7.5]) in being mixed-layer — substratum cubic-group structure combined with electroweak-symmetry-breaking emergent machinery.
- The mass ratio $m_b/m_\tau = 4.28/Z_S$ ([SM §7.5]) pending the explicit OI-vertex 1-loop computation confirming $K = 1/2$, together with four bridge gaps in the SχPT inheritance ([SM §7.5]): the substratum measure reduction under Theorem 2, the 3D MC vs 4D SχPT dimensional bridge, the 3D BZ vs 4D Dirac taste-structure equivalence, and the 3D vs 4D parity identification of chirality. The per-vertex $\cos^2$ symmetrization producing $K = 1/2$ is dimension-independent and structurally robust; the four bridges are derivation gaps inherited by the other §7 mass-cluster predictions that share the SχPT machinery. The closure path is the two-loop SχPT calculation specializing Panagopoulos–Spanoudes 2017 to the OI cubic-group setting.
- The PMNS predictions $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2)$ and $\sin^2\theta_{23} = 1/2 + 1/(2\pi^2)$ ([SM §7.3]) pending derivation of Cond 2 (the structural relation among A_2 Wilson coefficients) from cubic-lattice Yukawa structure. The reactor angle $\sin^2\theta_{13}$ is independent of Cond 2; it inherits the Cabibbo λ and is conditional with it (class C in [SM §7.6]; not counted separately in Class B).
- The MOND critical acceleration $a_0 = cH/6$ ([GR §7.3]), inheriting the conditions on the de Sitter temperature of [GR §3.2] and the coupling statement $Q = M_B c^2$ that the open G3 map would supply, with the 1/6 dimensional factor imported from Verlinde 2016 (eq. 1.7) as $(d-3)/[(d-2)(d-1)]$ in $d = 4$. Dependent predictions (baryonic Tully-Fisher, crossover radius r_M) inherit B-L status.

Class B-L violations at 3σ on precision upgrade are addressed under the pre-registration commitment via three branches: (a) closure of the open derivation *confirms* the prediction (recoverable from a transient data fluctuation); (b) clo-

sure *fails to confirm* the prediction while preserving the framework's structural commitments (recoverable error in the specific derivation, with the structural commitments intact); (c) demonstration that no consistent OI-internal derivation can produce the observed value (fatal, framework-falsifying).

Class B-M (mass chain). Two predictions inherit a single empirical input scale through a structural relation:

- The electron and muon masses m_e and m_μ via the Koide chain from the empirical input m_τ ([SM §7.2]). The prediction is structurally $\theta_0 = 2/9$ with $Q = 2/3$; the absolute scale is set by m_τ.
- The Higgs mass m_H via SM RGE running from the structural boundary condition $\lambda(M_{\rm Pl}) = 0$ ([SM §7.4]) with m_t as the empirical input.

Class B-M predictions falsify under the same precision-upgrade protocol as B-L, but the inheritance is from an empirical input scale rather than from an open derivation.

Class B-R (retrodictions). Two gauge-coupling values are explicitly retrodictions per the §6.3 [SM] parameter count:

- The SU(2) gauge coupling at M_Z ([SM §6, items 11-12 in §7.6]).
- The SU(3) gauge coupling at M_Z.

Three fitted parameters (δ_0, A, B) against three observed couplings produces zero residual by construction; these classifications are honest. Class B-R predictions can never falsify in the precision-upgrade sense — their value is what falsification of the *framework's gauge-sector structural commitments* (universality of δ_0, the resummation form, the structural prediction $1/\alpha_0 = 23.25$ from the 1-loop staggered VP (ambient coefficient; native for hypercharge only under H-observer-bundle and H-Y-vertex, [SM §6.5]), and the $A \cdot B$ cross-check (≈ 48 under the $1/N^2$ leading-power assumption; inconclusive across finite-size models, [SM §6.2.1])) would test.

Status summary. Of the predictions in Class B: **1 is structural with no open link of its own** (B-S), **2 are conditional structural** (B-C), **6 are layered conditional pending open derivations** (B-L), **2 are mass-chain inheritances** (B-M), and **2 are retrodictions** (B-R). Total 13 predictions in Class B. Their empirical standing is assessed entry by entry: the gauge couplings match by construction, and the quoted precision of a measured comparison does not apply to every entry.

The current status of each is documented in the SM and GR companion papers with experimental references. As experimental precision improves, the discrimination power of the B-S set increases monotonically; closure of the open derivations would move B-C and B-L predictions to B-S over time.

The Bekenstein-Hawking coefficient $S = A/(4l_p^2)$, with the factor 1/4 the ratio $2\pi/(8\pi)$ between the Euclidean KMS period and the Jacobson Einstein-equation prefactor ([GR §5]), is a consistency relation of the horizon calibration and is

not counted among them. It is conditional on the infrared detailed-balance condition H-slope on the horizon reservoir together with the horizon and frame conditions of [GR §2, §8.5]; the counting $S = \sigma A/\epsilon^2$, for an entropy σ per boundary mode, is unconditional and the coefficient enters through $\epsilon^2 = 4\sigma\, l_p^2$, from which σ cancels. The closure path is the continuum-generator theorem of [GR §2]. GW250114 confirms the classical area theorem, which the framework preserves; the 1/4 coefficient has no direct empirical test.

Class C: Framework-level commitments. Two broader commitments are implicit in the construction; each is stated with what would contradict it.

- *Finite classical realizability of quantum processes.* The characterization theorem ([Main, §3.4]) establishes that the quantum representation is the description admitted by an embedded observer satisfying C1–C4 on a deterministic substrate with bounded classical hidden-sector memory. At finite accuracy, every finite adaptive quantum experiment has a finite deterministic realization (the gluing and uniform-continuum theorems of [Main, §3.4]), while the exact continuum theory at any fixed finite carrier is excluded by the classical-dimension obstruction. Non-Markovian models are permitted and expected (they are how the hidden-sector correlation persistence manifests). No finite-accuracy experiment contradicts this commitment by itself. A memory-witness experiment would contradict the ontology only given the coherent-preparation and instrument bridge, an outcome-independent rule identifying fundamental open-system processes, and a constraint derived from the completed model on what those processes must show; the last is not established ([Main, §4.5]).
- *The dark sector as entropy-displacement, not as a novel particle content.* The framework's account of dark energy as boundary-entropy bookkeeping read in the Type II RVM channel, and dark matter as frozen boundary entropy (GR §7.3), is incompatible with the direct detection of a WIMP interpretable as a fundamental particle with Standard-Model-like interactions and no entropy-displacement signature. A confirmed WIMP with such properties falsifies the framework's dark-sector account while leaving the rest of the structure intact — it would require abandoning the §7.3 derivation and either identifying an error in it or accepting that the framework's dark-sector story is wrong even as other derivations survive.

Cosmological commitment. The dark energy sector is subtler than Classes A–C. The framework supplies geometry-dependent dark-energy bookkeeping whose smooth late-time expansion contains an $H^2 - H_0^2$ term — the local form shared by every smooth model — and reads it in the Type II running-vacuum channel; it does not presently derive a distinctive Type II law (GR §7.1). The magnitude is given by the emergent-QFT calculation $|\nu_{\mathrm{QFT}}| \sim (M_{\mathrm{SM}}/M_{\mathrm{Pl}})^2 \sim 10^{-32}$ (conditional on the GR §7.3 split, not yet established at theorem level), observationally indistinguishable from zero. Substratum contributions beyond this are constrained by the boundary mode count of §3-§5, which catalogues modes by spatial location alone (one mode per ϵ^2 cell, no inherent frequency

assignment — a count forced by the link-direction structure of the $K = 2d$ components, [GR §3.1]); the natural 2D distribution concentrates them near the UV cutoff, contributing well below $|\nu_{\mathrm{QFT}}|$. The framework's commitment is therefore the Type II reading, with magnitude $|\nu| \approx |\nu_{\mathrm{QFT}}|$ (conditional, per GR §7.1), and the Bertini et al. (2025) measurement $\nu = -(2.5 \pm 1.3) \times 10^{-4}$ consistent with this expectation at 2σ, where Bertini is itself consistent with zero. The falsification band is correspondingly stratified: a DESI Year 5 measurement of $|\nu| \gtrsim 10^{-4}$ at high significance would require additional substratum structure (log-uniform mode-per-decade distribution at the cosmological horizon) not currently in §3-§5; a measurement of ν with high-significance positive sign would falsify the Type II functional form itself, which is a structural commitment rather than a magnitude one. The first outcome is a refinement within the current bracket; the second is a structural falsification.

Unfixed inputs and falsification. The absolute mass scales, unfixed CP phases, initial conditions and other explicitly supplied model inputs are not parameter-free predictions. Once a model fixes them, it must still satisfy the empirical constraints and any structural relations claimed for that model. Reciprocity of transition counts fixes neither $\bar{\theta}$ nor the Jarlskog invariant [SM §5.3–§5.4]. The dynamics φ is not a microstate: changing it can change observable predictions. Theorem 24 proves specified transformations observationally invisible on specified carriers; it does not classify every choice of φ as gauge. A hidden initial state can also affect later observations. Only differences proved equivalent for the chosen preparation/readout carrier are empirically redundant relative to that carrier.

The pre-registration commitment. The author commits to treating the Class A conditions, the Class C commitments, and a Class B retrodiction exceeding 3σ on precision upgrade as framework-falsifying rather than as an invitation to post-hoc rescue. In the case of a Class B violation, the framework will be revisited to determine whether the violation reflects a computational error in the specific retrodiction, an incorrect specific-bijection input, or a structural problem with the derivation; the first two are recoverable, the third is fatal. The framework does not reserve the right to add structural postulates after the fact to accommodate disconfirming evidence. Pre-registration of this commitment is itself part of the falsifiability content of the framework.

Regimes of deliberate silence. Three regimes lie outside the framework's predictive scope by construction and are not objects of falsification claims. First, the *pre-horizon early universe* — the regime before a stable causal partition with $\tau_B \sim H^{-1}$ exists — has no observer in the framework's sense ([Main, §1.3]) and no horizon-level structure for the [GR] derivations to act on; the framework's only commitment in this regime is a boundary condition on deeper theories, namely that any successful pre-horizon dynamics must produce a C1–C4-satisfying regime by the time a horizon stabilizes. Second, the *equilibrium phase* of (S, φ) is, by the structural observer-selection theorem, observer-free for sufficiently fast-mixing equilibria ($\tau_{\mathrm{mix}} \ll \tau_S$) in the coarse-grained description

generated by iterating the one-step effective channel, conditional on its effective-mixing hypothesis ([Main, §4.6]); the framework makes no prediction for this regime because it has no observers to whom a prediction could be addressed. Third, regions *beyond the observer's causal horizon* are part of that observer's hidden sector and are addressed only through boundary effects (the area-law entropy of [GR, §5]); the framework takes no observer-independent position on their internal content. These silences are structural rather than methodological — they follow from the characterization theorem's identification of QM with embedded observation, and from the observer-selection theorem's restriction of observers to the non-equilibrium phase — and stating them explicitly sharpens the scope of the positive commitments above.

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7. Conclusion

Theorem 23 collects conditional local-branch consequences under the selected wave-rule, H-link, representation, amplitude-scale and matter premises, with the gravitational calibration retaining its horizon conditions. It does not prove the converse from E1–E7/A1–A6 to a unique local substratum. Theorem 24 establishes exact hidden-only channel invariance, broader rooted-population invariance and well-defined independent ancillary enlargement, while its alphabet/graph comparisons retain their structural scope. Pair C refutes exhaustiveness of the restricted coherent-channel moves. Proposition 24.2 separately gives fixed-carrier hidden-unitary rigidity from all block-word traces and the spanning premise; visible channel equality alone does not supply those hypotheses. The all-partitions classification, the full Standard Model gauge-image map and a common Bell-compatible physical completion remain open. The quantum, gauge and horizon constructions motivate a common-substratum interpretation, without establishing a unique full observational quotient.

The synthesis claim that follows is not a unification of quantum mechanics and general relativity in the traditional sense. The traditional unification problem treats the two theories as competing accounts of the same regime and asks how to merge them. The present framework treats them as projections of the same construction onto two different substructures — the visible-sector trace-out and the boundary-thermodynamic limit — and the question of how to merge them is dissolved as a category error rather than answered as a technical problem. The result is not a theory of everything but a structural account of why the apparent incompatibility between quantum mechanics and general relativity has been so resistant to resolution: there is nothing to resolve, because the two are not in competition. They are co-derived from the same object by the same procedure under the same conditions, and the framework developed across this four-paper synthesis ([Main], [SM], [GR], [Substratum]) makes this exact rather than approximate.

Three sets of open problems remain. Neither these results nor this four-paper

synthesis require the framework to be the final theory of physics: the substratum may itself be an effective description of something deeper, and the trace-out may have corrections beyond the leading-order results developed here. First, the *specific Bell-inclusive completion* that describes our universe is not uniquely determined by the framework. Theorem 23 gives conditional local lattice/gauge outputs; selection of a unique branch modulo a full gauge classification and the preparation-indexed Bell-composite rule remain open. The framework predicts the structural features of the Standard Model and the gravitational sector but not the specific values of (for example) the lightest fermion mass or the specific numerical value of the Type II RVM coefficient ν beyond its dimensional order of magnitude and sign. These are properties of the particular bijection, analogous to the mass of the sun in general relativity. Second, the *trans-Planckian regime* — the regime in which the lattice spacing $\epsilon = 2\sqrt{\sigma}\, l_p$ is not small compared to the relevant length scale — lies outside the framework's leading-order results. The framework presents the lattice as fundamental, not as a regulator approximating a continuum theory, and so does not need a continuum limit in the standard sense; but the corrections to the leading-order results in the regime where the partition geometry varies on lattice scales are an open question. Third, the *initial conditions* — the specific configuration of the bijection at any given time — are likewise not determined by the framework. The framework establishes which structures emerge from the bijection but not which microstate the universe is currently in.

These open problems are framed within the construction rather than against it. Each is sharply formulated and admits a definite (if presently unanswered) form. The progress reported here is that the structural relationship between quantum mechanics and general relativity — the central open problem of fundamental physics for nearly a century — is fixed by the construction developed across this four-paper synthesis, and the remaining open problems are problems within the construction rather than problems with it.

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